

**1997–2023 IMO Hong Kong Team  
Selection Test**

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## Problems

### Year 1997

#### Test 1

1. Prove that for positive real numbers  $x, y, z$ ,

$$\frac{xyz(x+y+z+\sqrt{x^2+y^2+z^2})}{(x^2+y^2+z^2)(xy+yz+zx)} \leq \frac{3+\sqrt{3}}{9}.$$

2.  $ABCDEF$  is a hexagon inscribed in a circle. Show that the diagonals  $AD, BE, CF$  are concurrent if and only if  $AB \cdot CD \cdot EF = BC \cdot DE \cdot FA$ .
3. Prove that there are infinitely many positive integers  $n$  such that  $2^n+1$  is divisible by  $n$ . Find all such  $n$ 's that are prime numbers.
4. Let  $x_1, x_2, x_3, \dots$  be a sequence of nonzero real numbers satisfying

$$x_n = \frac{x_{n-2}x_{n-1}}{2x_{n-2} - x_{n-1}} \quad \text{for } n = 3, 4, 5, \dots$$

Find all pairs  $(x_1, x_2)$  such that  $x_n$  is an integer for infinitely many  $n$ .

5. One may perform the following two operations on a positive integer:
- (a) multiply it by any positive integer; or
  - (b) delete zeros in its decimal representation.

Prove that for every positive integer  $X$ , one can perform a sequence of these operations that will transform  $X$  to a one-digit number.

#### Test 2

1. For a positive integer  $n$ , let  $f(n)$  be the largest  $k$  such that  $2^k$  divides  $n$  and  $g(n)$  be the sum of the digits of the binary representation of  $n$ . Prove that for any positive integers  $n$ ,
- (i)  $f(n!) = n - g(n)$ ;
  - (ii) 4 divides  $\binom{2n}{n} = \frac{(2n)!}{n!n!}$  if and only if  $n$  is not a power of 2.
2. Let the angle bisectors of  $\angle A, \angle B, \angle C$  of  $\triangle ABC$  intersect the circumcircle of  $\triangle ABC$  at  $P, Q, R$  respectively. Prove that  $AP + BQ + CR > BC + CA + AB$ .
3. Let  $\mathbb{Z}$  denote the set of integers. Find all functions  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  such that  $f(-1) = f(1)$  and  $f(x) + f(y) = f(x + 2xy) + f(y - 2xy)$  for all integers  $x, y$ .

4. Start with some positive integer. The following operation is performed on the number: its unit digit is split off and multiplied by 4, then this product is added to the remaining number. (For example, 1997 is changed to  $7 \times 4 + 199 = 227$ .) The operation is performed again and again. Prove that if the sequence of numbers obtained contains 1001, then none of the numbers in the sequence can be a prime number.

### Test 3

1. Find all positive integers which cannot be written as the sum of at least two consecutive positive integers.
2. Let  $f(x) = x^n + a_1x^{n-1} + \dots + a_n$  be a polynomial of degree  $n \geq 2$  with real roots  $b_1, b_2, \dots, b_n$ . If  $x$  is greater than  $b_1, b_2, \dots, b_n$ , prove that

$$f(x+1) \left( \frac{1}{x-b_1} + \frac{1}{x-b_2} + \dots + \frac{1}{x-b_n} \right) \geq 2n^2.$$

3. From a point  $P$  outside a circle centred at  $O$ , draw the two tangents to the circle touching it at  $A, B$ . Let  $M$  be a point on the segment  $AB$  and let  $C, D$  be points on the circle with midpoint  $M$ . Let the tangents to the circle at  $C, D$  intersect at  $Q$ . Show that  $OQ \perp PQ$ .
4. Five numbers 1, 2, 3, 4, 5 are written on a blackboard. A student may erase any two of the numbers  $a$  and  $b$  on the board and write the numbers  $a+b$  and  $ab$  replacing them. If this operation is repeatedly performed, can the numbers 21, 27, 64, 180, 540 ever appear on the board at the same time?

## Year 1998

### Test 1

1. Prove that  $S = \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n+1}$  is not an integer if  $n \geq 1$ .
2. If  $p$  is any prime other than 2 or 5, prove that  $p$  divides infinitely many of the integers  $9, 99, 999, 9999, \dots$
3.  $I$  is the incentre of  $\triangle ABC$ .  $P, Q, R$  are the circumcentres of the triangles  $IBC, ICA$  and  $IAB$  respectively. Prove that the points  $A, R, B, P, C, Q$  are concyclic.
4.  $ABC$  is a triangle;  $ABX, BCY$  and  $CAZ$  are similar isosceles triangles outside  $ABC$  such that  $XA = XB, YB = YC$  and  $ZC = ZA$ .  
Prove that  $AY, BZ$  and  $CX$  are concurrent.
5. Let  $a_i, b_i, c_i$  and  $R_i$  ( $i = 1, 2$ ) be the sides and the radii of the circumscribed circles of  $\triangle A_i B_i C_i$ . Prove that for  $m, n \geq 1$ ,  $a_1^m a_2^n + b_1^m b_2^n + c_1^m c_2^n \leq 9(2^{m+n-2} R_1^m R_2^n)$ .
6. (a) Given that  $(1+x)^{p-2} = 1 + a_1 x + a_2 x^2 + \cdots + a_{p-2} x^{p-2}$  where  $p$  is an odd prime number, prove that  $a_1 + 2, a_2 - 3, a_3 + 4, \dots, a_{p-3} - (p-2)$  and  $a_{p-2} + (p-1)$  are all multiples of  $p$ .  
(b) How many integer solutions are there to the equation  $x_1 + x_2 + x_3 + x_4 = 100$  where  $-1 \leq x_1 \leq 10, 0 \leq x_2 \leq 11, 1 \leq x_3 \leq 12$  and  $x_4 \geq 50$ ?

### Test 2

1. The general term of a sequence  $\{a_n\}$  is given by  $a_{n+2} = a_{n+1} - a_n$ . If the sum of the first 1997 terms is 1879 and the sum of the first 1879 terms is 1997, what is the sum of the first 2000 terms?
2. Show that  $\sqrt{a-1} + \sqrt{b-1} + \sqrt{c-1} \leq \sqrt{c(ab+1)}$  for real numbers  $a, b, c \geq 1$ .
3. Let  $f$  be a function with the following properties:
  - (i)  $f(n)$  is defined for every positive integer  $n$ ;
  - (ii)  $f(n)$  is a positive integer;
  - (iii)  $f(f(m) + f(n)) = m + n$  for all  $m$  and  $n$ .Find  $f(1997)$ .
4. Recall that  $n$  is perfect if the sum of the divisors of  $n$  is  $2n$ . Suppose now  $n$  is an odd perfect number. Show that  $n$  has at least 3 distinct prime factors.

5. There are  $n$  vertices on a plane. A graph  $G$  is formed by connecting every two vertices by exactly one edge, and painting all edges, each in one colour, chosen from  $k$  given colours.
- (a) For  $n = 6$  and  $k = 3$ , show that it is possible to paint the edges so that the graph contains no closed circuit painted in one colour only.
  - (b) Prove that for every positive integral value of  $k$  (i.e. number of colours allowed), if the number of vertices  $n \geq 2k + 1$ , then it is always possible to find a mono-coloured closed circuit in  $G$ .
6. The diagonals of a quadrilateral  $ABCD$  intersect at  $K$ .  $M, N$  are the midpoints of  $AB$  and  $CD$  respectively. Let  $k, m, n$  be the lines through  $K, M, N$  perpendicular to  $AD, BD$  and  $AC$  respectively. Show that if  $k, m, n$  are concurrent, then  $A, B, C, D$  are concyclic.

## Year 1999

### Test 1

- Let  $ABCD$  be a trapezoid with  $AB \parallel CD$ ,  $AB > CD$  and  $AD > BC$ . Show that
  - $\angle CBA > \angle DAB$ ,
  - $AC > BD$ .
- Let  $a_1, a_2, \dots, a_n > 0$ . Prove that  $a_1^{a_1} a_2^{a_2} \cdots a_n^{a_n} \geq (a_1 a_2 \cdots a_n)^{\frac{a_1 + a_2 + \cdots + a_n}{n}}$ .
  - Let  $x, y, z \geq 0$  and  $x + y + z = 1$ . Prove that  $0 \leq xy + yz + zx - 3xyz \leq \frac{1}{4}$ .
- Suppose  $p$  and  $q$  are positive integers such that  $\frac{p}{q} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{1335}$ . Show that 2003 is a factor of  $p$ .
- Find all polynomials  $f(x)$  such that  $f(f(x)) = (f(x))^m$ , where  $m > 1$  is a fixed integer. Substantiate your answer.
- Suppose there are 6 members in an International Mathematical Olympiad team. Prove that among these 6 members there are three members who either all know each other or all don't know each other.
- $k$  cards numbered 1 to  $k$  are arranged at random in a row. In a move, we may change any block of consecutive cards whose numbers are in ascending or descending order and switch the block around. For example, in the case  $k = 4$ , 4132 may be changed to 1432 and 3214 may be changed to 1234. Suppose that in at most  $n_k$  moves, we can arrange the  $k$  cards so that their numbers are in ascending or descending order.
  - Prove that  $n_4 = 3$ .
  - Determine or estimate  $n_5$ .
- In this problem,  $G$  denotes a connected graph without parallel edges (but not necessarily a planar graph) with complement  $\overline{G}$ . Let  $V$  be the set of vertices,  $E$  be the set of edges, and  $\Delta(G)$  or  $\Delta_G$  be the maximum degree of the vertices. Define the *chromatic number*  $\chi(G)$  or  $\chi_G$  by

$$\chi(G) = \chi_G = \min \{k : G \text{ has a } k\text{-colouring}\}.$$

In other words,  $\chi(G)$  is the minimum number of colours needed to colour the vertices of  $G$  such that no two adjacent vertices have the same colour.

- Show that if every vertex of  $G$  has degree 2, then  $V$  and  $E$  form the corners and sides of a polygon. These are called *cycle graphs*.
- Prove that  $\chi_G \leq \Delta_G + 1$ .
- Give two types of examples that  $\chi_G = \Delta_G + 1$ .
- Show that if  $G$  is not any of the above types, then  $\chi_G \leq \Delta_G$ .
- Give an example of  $G$  such that  $\chi(G) + \chi(\overline{G}) = \#V + 1$ .

## Year 2000

### Test 1

1. Find the remainder of  $1999^{2000}$  when it is divided by 31.
2. Let  $a, b, c$  be positive and  $abc = 1$ . Prove that

$$\frac{1+ab^2}{c^3} + \frac{1+bc^2}{a^3} + \frac{1+ca^2}{b^3} \geq \frac{18}{a^3+b^3+c^3}.$$

Determine when equality is attained.

3. On a table there are 1999 tea cups with their mouths facing upward initially. In each move, 100 of them are turned upside down. After a number of moves, can they be turned so that all their mouths face downward? Why?  
Answer the above two questions for the case where the number of cups is 1998.
4.  $ABC$  is a triangle with  $BC > CA > AB$ .  $D$  is a point on side  $BC$ , and  $E$  is a point on  $BA$  produced beyond  $A$  so that  $BD = BE = CA$ . Let  $P$  be a point on side  $AC$  such that  $E, B, D, P$  are concyclic, and let  $Q$  be the second intersection point of  $BP$  with the circumcircle of  $\triangle ABC$ . Prove that  $AQ + CQ = BP$ .
5. Let  $G$  be a connected simple graph which has  $2p$  vertices (where  $p$  is a positive integer), but contains no triangles. Prove that the number  $\#E$  of its edges satisfies  $\#E \leq p^2$ .
6. Determine all primes of the form  $n^n + 1$ , which are less than  $10^{19}$  ( $n$  is a positive integer).

### Test 2

1. Prove that for any integer  $n$ ,  $n^{30} - n^{14} - n^{18} + n^2$  is divisible by 46410.
2. (a) Let  $a, n, k$  be positive integers. Prove that  $a^{4n+k}$  and  $a^k$  have identical first (rightmost) digits in their decimal representations.  
(b) Find the first digit of the decimal representation of the number

$$2^{1999} + 7^{1999} + 9^{1999}.$$

3. Let  $ABC$  be an acute triangle. Suppose a circle  $\Gamma_1$ , with centre  $O_1$ , touches the sides  $BC$  produced at  $E$ ,  $AC$  produced at  $G$ , and  $AB$  at  $C'$ . Suppose also that another circle  $\Gamma_2$ , with centre  $O_2$ , touches the sides  $AB$  produced at  $H$ ,  $BC$  produced at  $F$ , and  $AC$  at  $B'$ . Let the extensions of  $EG$  and  $FH$  intersect at  $P$ . Prove that  $PA \perp BC$ .
4. Let  $a, b, c, d, \ell, m, n, r$  be given (fixed) positive numbers. Suppose  $x, y, z, w$  are positive numbers satisfying  $ax + by + cz + dw = 2$ .

Determine the minimum of  $F = \frac{\ell}{x} + \frac{m}{y} + \frac{n}{z} + \frac{r}{w}$ .



5. For a graph  $G$ ,

$$\chi_G = \min \{k : G \text{ has a } k \text{ colouring}\},$$
$$\Delta_G = \max \{\text{degrees of vertices of } G\}.$$

- (a) Prove that for any simple connected graph  $G$ ,  $\chi_G \leq \Delta_G + 1$ .
- (b) A regular graph is a graph that every vertex has the same degree. Prove that if a simple connected graph  $G$  is not regular, then  $\chi_G \leq \Delta_G$ .

**Year 2001**

Test 1

1. Let  $\triangle ABC$  be an acute triangle. Let  $O$  be the centre of a circle which passes through  $A, C$  and meet  $AB, BC$  at  $K, N$  respectively (where  $K$  and  $N$  are distinct). Suppose the circumcircles of  $\triangle ABC$  and  $\triangle BKN$  meet again at  $M$  (a point other than  $B$ ). Prove that  $\angle OMB$  is a right angle.
2. Find all positive integers  $n$  for which the number  $n^8 - n^2$  is not divisible by 72.
3. For any positive integer  $n$ , determine (with proof) if the polynomial

$$f(x) = (x^2 + x)^{2^n} + 1$$

can be factored into a product of two non-constant polynomials with integer coefficients.

4. Let  $a, b, c$  be positive real numbers. Prove that

$$(a+b)^2 + (a+b+4c)^2 \geq \frac{100abc}{a+b+c}.$$

5. Let  $S = \{1, 2, \dots, 120\}$ ,  $M$  a subset of  $S$ . Suppose it is impossible to find 8 members from  $M$  whose greatest common divisor exceeds 1. Determine the largest possible value of  $|M|$ .

## Year 2002

### Test 1

1. Find all integers  $n$  satisfying all three conditions

$$n \equiv 2 \pmod{3}, \quad n \equiv -1 \pmod{5} \quad \text{and} \quad n \equiv 3 \pmod{7}.$$

2. (a) Using the characteristic equation method or otherwise, find a formula in terms of  $n$  for the sequence  $a_0 = 1$ ,  $a_1 = 4$  and  $a_{n+2} = 5a_{n+1} - 6a_n$  for  $n = 0, 1, 2, \dots$
- (b) Using the characteristic equation method or otherwise, find a formula in terms of  $n$  for the sequence  $b_0 = 3$ ,  $b_1 = 7$  and  $b_{n+2} = 4b_{n+1} - 4b_n$  for  $n = 0, 1, 2, \dots$
3. Prove that if  $a, b, c$  are positive real numbers, then

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

4.  $\triangle ABC$  is isosceles with  $AB = AC$ .  $D$  is a point on the minor arc  $AC$  on the circumcircle of  $\triangle ABC$ . Let  $E$  be the foot of the perpendicular from  $A$  to  $BD$ . Prove that  $BE = ED + DC$ .
5. Considering  $(\text{mod } 4)$  or otherwise, show that if  $a, b, c, d$  are integers such that

$$(3a + 5b)(7b + 11c)(13c + 17d)(19d + 23a) = 2001^{2001},$$

then  $a$  must be even.

6.  $\triangle ABC$  is an acute triangle. Let  $A_1$  be the centre of the square inscribed in  $\triangle ABC$  having two vertices on side  $BC$ . Let  $B_1$  be the centre of the square inscribed in  $\triangle ABC$  having two vertices on side  $CA$ . Let  $C_1$  be the centre of the square inscribed in  $\triangle ABC$  having two vertices on side  $AB$ . Prove that lines  $AA_1, BB_1$  and  $CC_1$  are concurrent.
7. For an irrational number  $x$ , let  $x'$  be the integer nearest to  $x$ . Define  $\langle x \rangle = |x - x'|$ . Show that for every irrational number  $y$ , the minimum of the numbers  $\langle y \rangle, \langle 2y \rangle, \dots, \langle 2001y \rangle$  is less than  $\frac{1}{2001}$ .

## Year 2003

### Test 1

1. Let  $a, b$  and  $c$  be nonnegative real numbers. Prove that

$$\frac{(a+b+c)^2}{3} \geq a\sqrt{bc} + b\sqrt{ca} + c\sqrt{ab}.$$

2. Let  $\triangle ABC$  be a triangle.  $M$  is the midpoint of  $AC$ ,  $D$  is a point on  $AB$ .  $BM$  and  $CD$  meet at  $O$ , with  $AB = CO$ .  $E$  is a point on  $AC$  such that  $DE \parallel BM$ . Prove that  $AB \perp BC$  if and only if  $ADOM$  is a cyclic quadrilateral.
3. Let  $n$  be an integer greater than 1. In a school there are  $n^2 - n + 2$  clubs and each club has exactly  $n$  members. Each pair of clubs has exactly one member in common. Show that there is one student belonging to all of the clubs.
4. Show that there exists a positive integer  $k$  such that  $k \times 22^n + 1$  is not prime (composite) for every positive integer  $n$ .
5. Denote by  $\sigma(n)$  the sum of divisors of  $n$ . A positive integer  $n$  ( $n \geq 2$ ) is redundant if for any integer  $k$ , with  $k < n$ , we have  $\frac{\sigma(k)}{k} < \frac{\sigma(n)}{n}$ . (For instance if  $n = 1, 2, 3, 4, 5$ , then  $\frac{\sigma(n)}{n} = 1, \frac{3}{2}, \frac{4}{3}, \frac{7}{4}, \frac{6}{5}$  respectively. Hence, 2 and 4 are redundant, 3 and 5 are not.) Show that there exist infinitely many redundant numbers.
6. In a conference, there are 2002 representatives from 100 countries. The number of representatives from each country is at least 1 and at most 45. They are seated in rows with each row consisting of 45 seats. It is required that the representatives from the same country must be seated in the same row. What is the smallest number of rows needed to ensure that all representatives can be seated? Justify your answer.
7. Let  $n \geq 3$  be an integer and  $x_1, x_2, \dots, x_{n-1}$  be nonnegative integers such that
- (i)  $x_1 + x_2 + \dots + x_{n-1} = n$ ,
  - (ii)  $x_1 + 2x_2 + \dots + (n-1)x_{n-1} = 2n - 2$ .

Find the minimum of the sum  $\sum_{k=1}^{n-1} kx_k(2n-k)$ . Justify your answer.

## Year 2004

### Test 1

1. Given a set  $S$  of 10 distinct positive integers, all less than 118, prove that  $S$  contains at least two nonempty subsets  $A$  and  $B$ , such that  $A \cap B = \emptyset$ , and the sums of the integers in  $A$  and  $B$  are equal.
2. Let  $a, b, c$  be positive real numbers. Prove that  $\frac{a^4}{bc} + \frac{b^4}{ca} + \frac{c^4}{ab} \geq a^2 + b^2 + c^2$ .
3. A positive integer  $n$  is said to be *good* if there exist positive integers  $a$  and  $b$  such that  $a + b = n$  and  $ab$  is divisible by  $n$ . For instance, 9 is good since  $3 + 6 = 9$  and  $3 \times 6$  is divisible by 9. Furthermore,  $n$  is said to be *very good* if we can choose  $a$  and  $b$  to be distinct. Hence 9 is very good.
  - (a) Given that 2003 is a prime, show that 2003 is not good.
  - (b) Is 2002 good?
  - (c) Find all positive integers which are good but not very good. Justify your answers.
4. Let  $ABCD$  be a quadrilateral. Let  $P$  be the intersection of  $AC$  and  $BD$ . Let  $M, N$  be the midpoints of  $AB, CD$ , respectively. Prove that the lines through  $P, M, N$  perpendicular to  $AD, BD, AC$ , respectively, are concurrent if and only if  $ABCD$  is cyclic.
5. For every positive integer  $n$ , let  $f(n)$  be the number of all positive integers with exactly  $2n$  digits, each having exactly  $n$  of the digits equal to 1 and the other  $n$  digits equal to 2. Let  $g(n)$  be the number of all positive integers with exactly  $n$  digits, each of its digits can only be 1, 2, 3 or 4 and the number of 1's equals the number of 2's. Prove that  $f(n) = g(n)$ .
6. Let  $p$  be a prime number and  $n$  a positive integer such that  $n \geq p$ . Show that  $\binom{n}{p} - \left\lfloor \frac{n}{p} \right\rfloor$  is divisible by  $p$ . (Here the combinatorial symbol  $\binom{n}{p} = \frac{n!}{p!(n-p)!}$  and  $[x]$  stands for the largest integer less than or equal to  $x$ .)

## Year 2005

### Test 1

1. Find, with reasons, all functions  $f : \mathbb{N} \rightarrow \mathbb{N}$  such that

- (i)  $f(n) = 1$  if and only if  $n = 1$ ,
- (ii) if  $d = \gcd(n, m)$ , then  $f(nm) = \frac{f(n)f(m)}{f(d)}$ , and
- (iii)  $f^{2004}(n) = n$  for every  $n \in \mathbb{N}$ , where  $f^{2004}(n) = \underbrace{f(f(\cdots f(n)))}_{2004}$ ,

or show that they do not exist.

2. Let  $O$  and  $H$  be respectively the circumcentre and orthocentre of  $\triangle ABC$ . Let  $A', B'$  and  $C'$  be the midpoints of  $BC, CA$  and  $AB$  respectively and  $D, E$  and  $F$  be respectively the feet from  $A, B$  and  $C$  to the opposite sides. Show that  $OA' \cdot HD = OB' \cdot HE = OC' \cdot HF$ .
3. Given that  $p, q$  and  $r$  are positive real numbers, prove that

$$\frac{1}{q+r} + \frac{1}{r+p} + \frac{1}{p+q} \geq \frac{9}{2(p+q+r)}.$$

Hence prove that if  $m$  is a real number greater than 1, then

$$\frac{p^m}{q+r} + \frac{q^m}{r+p} + \frac{r^m}{p+q} \geq \frac{(p+q+r)^{m-1}}{2 \cdot 3^{m-2}}.$$

4. Let  $AD, BE$  and  $CF$  be respectively the altitudes of  $\triangle ABC$ . Let  $P, Q$  and  $R$  be points on the lines  $BC, CA$  and  $AB$  respectively such that  $AP$  is perpendicular to  $EF$ ,  $BQ$  is perpendicular to  $FD$  and  $CR$  is perpendicular to  $DE$ . Show that  $AP, BQ$  and  $CR$  are concurrent.
5. For every positive integer  $n$ , find (with proof) the number of ways of sitting  $2n$  people  $A_1, B_1, A_2, B_2, \dots, A_n, B_n$  around a round table so that  $A_i$  and  $B_i$  are not next to each other for  $i = 1, 2, \dots, n$ . Two sitting arrangements are considered the same if one can be obtained from another by a rotation. (You may express your answer in whatever ways.)
6. Let  $r$  be the positive root of the equation  $x^2 - 2004x - 1 = 0$ . Define the sequence  $\{a_n\}$  as follows:

$$a_0 = 1, a_{n+1} = [ra_n], n \geq 0,$$

where  $[y]$  denotes the greatest integer not exceeding  $y$ . Find the remainder when  $a_{2004}$  is divided by 2004.

## Test 2

1. Let  $ABC$  be an acute triangle,  $D$  the foot of the perpendicular from  $A$  to  $BC$ . With  $AD$  as a diameter, draw a circle intersecting  $AB$  and  $AC$  at  $E$  and  $F$  respectively. Suppose  $AD$  and  $EF$  intersect at  $G$ , and  $AD$  is extended to meet the circumcircle of  $\triangle ABC$  at  $H$ . Show that  $AD^2 = AG \cdot AH$ .
2. Let  $M$  be the intersection point of the diagonals  $AC$  and  $BD$  of a convex quadrilateral  $ABCD$ . The bisector of  $\angle ACD$  meets  $BA$  extended at  $K$ . If  $MA \cdot MC + MA \cdot CD = MB \cdot MD$ , show that  $\angle BKC = \angle CDB$ .
3. Let  $x$  be the area of two non-congruent triangles with integral side lengths. Determine the smallest possible value of  $x$ . Justify your claim.
4. Suppose  $M$  is a point on the side  $AB$  of  $\triangle ABC$  such that the incircles of  $\triangle AMC$  and  $\triangle BMC$  have the same radius. The two incircles, centered at  $O_1$  and  $O_2$ , meet  $AB$  at  $P$  and  $Q$  respectively. It is known that the area of  $\triangle ABC$  is six times the area of the quadrilateral  $PQO_2O_1$ , determine the possible values of  $\frac{AC + BC}{AB}$ . Justify your claim.

## Year 2006

### Test 1

- Find the integral solutions of the equation  $7(x + y) = 3(x^2 - xy + y^2)$ .
- The function  $f(x, y)$ , defined on the set of all nonnegative integers, satisfies
  - $f(0, y) = y + 1$ ,
  - $f(x + 1, 0) = f(x, 1)$ , and
  - $f(x + 1, y + 1) = f(x, f(x + 1, y))$ .Find (a)  $f(3, 2005)$ , and (b)  $f(4, 2005)$ .
- In triangle  $ABC$ , the altitude, angle bisector and median from  $C$  divide the angle  $\angle C$  into four equal angles. Find  $\angle B$ .
- Let  $x, y, z$  be positive real numbers such that  $x + y + z = 1$ . For positive integer  $n$ , define  $S_n = x^n + y^n + z^n$ . Furthermore, let  $P = S_2 S_{2005}$  and  $Q = S_3 S_{2004}$ .
  - Find the smallest possible value of  $Q$ .
  - If  $x, y, z$  are pairwise distinct, determine whether  $P$  or  $Q$  is larger.
- Given finitely many points in a plane, it is known that the area of the triangle formed by any three points of the set is less than 1. Show that all points of the set lie inside or on the boundary of a triangle with area less than 4.
- Find  $2^{2006}$  positive integers satisfying the following conditions.
  - Each positive integer has  $2^{2005}$  digits.
  - Each positive integer only has 7 or 8 in its digits.
  - Among any two chosen integers, at most half of their corresponding digits are the same.

### Test 2

- Let  $ABCD$  be a cyclic quadrilateral. Show that the orthocentres of the triangles  $\triangle ABC$ ,  $\triangle BCD$ ,  $\triangle CDA$  and  $\triangle DAB$  are the vertices of a quadrilateral congruent to  $ABCD$  and show that the centroids of the same triangles are the vertices of a cyclic quadrilateral.
- $ABCD$  is a cyclic quadrilateral with  $BC = CD$ . The diagonals  $AC$  and  $BD$  intersect at  $E$ . Let  $X, Y, Z$  and  $W$  be the incentres of triangles  $\triangle ABE$ ,  $\triangle ADE$ ,  $\triangle ABC$  and  $\triangle ADC$  respectively. Show that  $X, Y, Z$  and  $W$  are concyclic if and only if  $AB = AD$ .
- $A$  and  $B$  are fixed points on a plane and  $L$  is a line passing through  $A$  and not  $B$ .  $C$  is a variable point moving from  $A$  toward infinity along a half-line of  $L$ . The incircle of  $\triangle ABC$  touches  $BC$  at  $D$  and  $AC$  at  $E$ . Show that line  $DE$  passes through a fixed point.



4. Let  $\triangle ABC$  be given with  $R$  radius of its circumcircle. Let  $AB = c$ ,  $BC = a$  and  $CA = b$ . Let  $K_1$  and  $K_2$  be the circles which pass through  $C$  and are tangent to  $AB$  at  $A$  and  $B$  respectively. Let  $K$  be the circle of radius  $r$  which is externally tangent to  $K_1, K_2$  and tangent to the line  $AB$ .
- (a) Express  $r$  in terms of  $a, b, c$  and  $R$ .
- (b) Suppose  $r = \frac{R}{4}$ . Find  $\angle C$ .

## Year 2007

### Test 1

1. Let  $p, q, r$  and  $s$  be real numbers such that  $p^2 + q^2 + r^2 - s^2 + 4 = 0$ . Find the maximum value of  $3p + 2q + r - 4|s|$ .
2. Let  $A, B$  and  $C$  be real numbers such that
  - (i)  $\sin A \cos B + |\cos A \sin B| = \sin A |\cos A| + |\sin B| \cos B$ ,
  - (ii)  $\tan C$  and  $\cot C$  are defined.

Find the minimum value of  $(\tan C - \sin A)^2 + (\cot C - \cos B)^2$ .

3. Let  $ABC$  be a triangle with  $AB = c$ ,  $BC = a$  and  $CA = b$ . Show that

$$\frac{\cos A}{a^3} + \frac{\cos B}{b^3} + \frac{\cos C}{c^3} \geq \frac{3}{2abc}.$$

4. (a) Given five points on a plane such that no three of the points are collinear, show that among the triangles which are drawn using any three of these five points as vertices, at least three of the triangles formed are not acute-angled triangles. (An acute-angled triangle is one in which all the three interior angles are acute angles.)  
(b) Given any 100 points on a plane such that no three of the points are collinear, show that among the triangles which are drawn using any three of these 100 points as vertices, at least 30% of the triangles are not acute-angled triangles.
5. The sequence  $\{a_n\}$  is defined by  $a_1 = 0$  and  $(n+1)^3 a_{n+1} = 2n^2(2n+1)a_n + 2(3n+1)$  for all integers  $\geq 1$ . Show that infinitely many members of the sequence are positive integers.
6. Determine all pairs  $(x, y)$  of positive integers such that  $\frac{x^2y + x + y}{xy^2 + y + 11}$  are integers.

### Test 2

1. Let  $ABC$  be an equilateral triangle with circumcentre  $O$ . Let  $P$  be a point inside the triangle such that  $\angle BPC = 120^\circ$  and  $P \neq O$ . Let  $B'$  and  $C'$  be the images of  $B$  and  $C$  respectively under the reflection in the line  $PO$ . Furthermore  $B'C'$  meets  $AP$  at  $D$ . Determine  $AD : DP$ . Justify your claim.
2. Let  $ABCD$  be a cyclic quadrilateral such that  $AD + BC = AB$ . Show that the intersection of the bisectors of  $\angle ADC$  and  $\angle BCD$  is on line  $AB$ .
3. Let  $ABCDEF$  be a convex hexagon such that  $AB \parallel DE$ ,  $BC \parallel EF$  and  $CD \parallel FA$ . Let  $H, I, J, K, L$  and  $M$  be the midpoints of sides  $AB, BC, CD, DE, EF$  and  $FA$ , respectively. Prove that lines  $HK, IL$  and  $JM$  are concurrent.

4. The incircle (with centre  $O$ ) of an isosceles triangle  $ABC$  with  $AB = AC$  meets  $BC, CA, AB$  at  $K, L, M$  respectively. Let  $N$  be the intersection of the lines  $OL$  and  $KM$ , and let  $Q$  be the intersection of the lines  $BN$  and  $CA$ . Let  $P$  be the foot of the perpendicular from  $A$  to  $BQ$ . Suppose  $BP = AP + 2PQ$ . Determine the possible value(s) of  $\frac{AB}{BC}$ .

## Year 2008

### Test 1

1. Let  $x$  and  $y$  be real numbers satisfying  $x - y + 1 \neq 0$ . If

$$1 + \cos^2(2007x + 2008y - 1) = \frac{x^2 + y^2 + 2(1+x)(1-y)}{x - y + 1},$$

find the minimum value of  $xy$ .

2. Show that the product of three consecutive integers is the sum of four integer cubes.
3. Let  $a, b$  and  $c$  be the sides of a triangle and  $p = a + b + c$ . Show that

(a)  $\frac{a}{b+c-a} + \frac{b}{c+a-b} + \frac{c}{a+b-c} \geq 3$ ;

(b)  $a^k + b^k > \frac{c^k}{2^{k-1}}$  for  $k = 2, 3, 4, \dots$ ;

(c)  $a^k + b^k + c^k \geq \frac{p^k}{3^{k-1}}$  for  $k = 2, 3, 4, \dots$

4. Let  $H$  be an arbitrary point on the altitude  $CP$  of an acute triangle  $ABC$ . The lines  $AH$  and  $BH$  intersect  $BC$  and  $AC$  in  $M$  and  $N$  respectively. Show that  $\angle NPC = \angle MPC$ .
5. Let  $x_1$  and  $y_1$  be positive integers satisfying  $x_1 + y_1 = 2^r$ , where  $r$  is a positive integer. If  $x_1 < y_1$ , let  $x_2 = 2x_1$  and  $y_2 = 2^r - x_2$ . On the other hand, if  $x_1 > y_1$ , then we let  $y_2 = 2y_1$  and  $x_2 = 2^r - y_2$ . Apply the same procedure to  $(x_2, y_2)$ , and so on. Show that the procedure will end in finitely many steps, i.e. there exists a  $k$  such that  $x_k = y_k$ . (For example, let  $(x_1, y_1) = (7, 57)$ , with  $x_1 + y_1 = 2^6$ . Then  $(x_2, y_2) = (14, 50)$ ,  $(x_3, y_3) = (28, 36)$ ,  $(x_4, y_4) = (56, 8)$ ,  $(x_5, y_5) = (48, 16)$ , and finally  $(x_6, y_6) = (32, 32)$ .)
6. For each permutation  $x_1, x_2, \dots, x_{10}$  of  $1, 2, \dots, 10$ , compute

$$|2x_1 - 3x_2| + |2x_2 - 3x_3| + |2x_3 - 3x_4| + \dots + |2x_{10} - 3x_1|.$$

Let  $S$  be the maximum possible value of this sum. Find the number of permutations attaining  $S$ .

### Test 2

1. Let  $p$  be a prime number and  $f(x)$  be a polynomial with rational coefficients and is of degree  $p$ . Furthermore (i)  $f(m)$  is an integer for any integer  $m$  and (ii)  $p$  divides  $f(p+1) - f(1)$ . Find all such polynomials.

2. Show that for any triangle  $ABC$  with area  $S$  and circumradius  $R$ ,

$$(a) \quad \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \leq \frac{3\sqrt{3}}{4S} + \left(\frac{1}{a} - \frac{1}{b}\right)^2 + \left(\frac{1}{b} - \frac{1}{c}\right)^2 + \left(\frac{1}{c} - \frac{1}{a}\right)^2,$$

$$(b) \quad \tan \frac{A}{2} + \tan \frac{B}{2} + \tan \frac{C}{2} \leq \frac{9R^2}{4S}.$$

Here  $AB = c$ ,  $BC = a$  and  $CA = b$ .

3. Show that  $\cos \frac{\pi}{7}$  is not of the form  $p + \sqrt{q} + \sqrt[3]{r}$ , where  $p, q$  and  $r$  are rational numbers.

4. Let  $\theta$  be a fixed angle. Let two circles touch each other internally at point  $A$ . Let  $AB$  be a chord of the outer circle and such that  $AB$  is not a diameter. Let  $M$  be a variable point on the major arc  $AB$  of the outer circle.  $MA$  meets the inner circle at  $N$ .  $P$  is a point on the segment  $MB$  such that  $\angle MPN = \theta$ . Determine the locus of  $P$  while  $M$  varies.

## Year 2009

### Test 1

1. Let  $\theta_1, \theta_2, \dots, \theta_{2008}$  be real numbers. Find the maximum value of

$$\sin \theta_1 \cos \theta_2 + \sin \theta_2 \cos \theta_3 + \dots + \sin \theta_{2007} \cos \theta_{2008} + \sin \theta_{2008} \cos \theta_1.$$

2. Find the total number of solutions to the following system of equations:

$$\begin{cases} a^2 + bc \equiv a \pmod{37}, \\ b(a + d) \equiv b \pmod{37}, \\ c(a + d) \equiv c \pmod{37}, \\ bc + d^2 \equiv d \pmod{37}, \\ ad - bc \equiv 1 \pmod{37}. \end{cases}$$

3. Let  $ABCDE$  be an arbitrary convex pentagon. Suppose that  $BD \cap CE = A'$ ,  $CE \cap DA = B'$ ,  $DA \cap EB = C'$ ,  $EB \cap AC = D'$  and  $AC \cap BD = E'$ . Suppose also that  $\odot ABD' \cap \odot AC'E = A''$ ,  $\odot BCE' \cap \odot BD'A = B''$ ,  $\odot CDA' \cap \odot CE'B = C''$ ,  $\odot DEB' \cap \odot DA'C = D''$  and  $\odot EAC' \cap \odot EB'D = E''$ . Prove that  $AA''$ ,  $BB''$ ,  $CC''$ ,  $DD''$  and  $EE''$  are concurrent. (Here  $\ell_1 \cap \ell_2 = P$  means that  $P$  is the intersection of lines  $\ell_1$  and  $\ell_2$ . Also  $\odot A_1A_2A_3 \cap \odot B_1B_2B_3 = Q$  means that  $Q$  is the intersection of the circumcircles of  $\triangle A_1A_2A_3$  and  $\triangle B_1B_2B_3$ .)
4. In a school there are 2008 students. Students are members of certain committees. A committee has at most 1004 members and every two students join a common committee.
- (a) Determine the smallest possible number of committees in the school.
- (b) If it is further required that the union of any two committees consists of at most 1800 students, will your answer in (a) still hold?
5. Let  $a, b, c$  be the three sides of a triangle. Determine all possible values of  $\frac{a^2 + b^2 + c^2}{ab + bc + ca}$ .
6. Show that the equation  $y^{37} \equiv x^3 + 11 \pmod{p}$  is solvable for every prime  $p$ , where  $p \leq 100$ .

### Test 2

1. Let  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  ( $\mathbb{Z}$  is the set of integers) be such that  $f(1) = 1$ ,  $f(2) = 20$ ,  $f(-4) = -4$  and  $f(x + y) = f(x) + f(y) + axy(x + y) + bxy + c(x + y) + 4$  for all  $x, y \in \mathbb{Z}$ , where  $a, b$  and  $c$  are certain constants.
- (a) Find a formula for  $f(x)$ , where  $x$  is any integer.
- (b) If  $f(x) \geq mx^2 + (5m + 1)x + 4m$  for all nonnegative integers  $x$ , find the greatest possible value of  $m$ .

2. Define a *k-clique* to be a set of  $k$  people such that every pair of them know each other (knowing is mutual). At a certain party, there are two or more 3-cliques, but no 5-clique. Every pair of 3-cliques has at least one person in common. Prove that there exists at least one, and not more than two persons at the party, whose departure (or simultaneous departure) leaves no 3-clique remaining.
3. Prove that there are infinitely many primes  $p$  such that  $N_p = p^2$ , where  $N_p$  is the total number of solutions mod  $p$  to the equation  $3x^3 + 4y^3 + 5z^3 - y^4z \equiv 0$ .
4. Two circles  $C_1, C_2$  with different radii are given in the plane. They touch each other externally at  $T$ . Consider any points  $A \in C_1$  and  $B \in C_2$ , both different from  $T$ , such that  $\angle ATB = 90^\circ$ .
  - (a) Show that all such lines  $AB$  are concurrent.
  - (b) Find the locus of midpoints of all such segments  $AB$ .

## Year 2010

### Test 1

1. In a chess tournament there were  $n$  ( $n > 10$ ) participants. Each participant plays against another exactly once. If a game ends in a draw, each participant gets 1 point, otherwise the winner gets 2 points and the loser gets none. At the end of the tournament, every participant found that half of his/her score came from games played against the ten participants with the lowest scores. This is true even for each of the ten participants with the lowest scores, (i.e. half of the scores of each of the 10 lowest scorers comes from playing with other 9 lowest scorers). Find the possible value(s) of  $n$ .
2. Let  $a, b, c$  be positive real numbers such that  $abc = 1$ . Show that

$$\frac{1}{a^2(b+c)} + \frac{1}{b^2(c+a)} + \frac{1}{c^2(a+b)} \geq \frac{3}{2}.$$

3. Let  $x^2 - p_i x + q_i = 0$ ,  $i = 1, 2, \dots, 15$  be 15 quadratic equations, where each of the  $p_i$  and  $q_i$ ,  $i = 1, 2, \dots, 15$  is one of the numbers  $1, 2, \dots, 30$ , with no repetition nor omission. Determine the maximum possible number of real roots among the equations which are of values greater than 20.
4. A rectangle with sides of integral lengths can be divided into two or more squares with sides of integral lengths and such that there is exactly one square with smallest side length. Determine the smallest possible area of the rectangle.
5. Let  $n$  be a positive integer. When  $\frac{n}{567}$  is expressed as a decimal, it is a recurring decimal with smallest period  $k$ . Find the sum of all possible values of  $k$ . (Note: We say that  $k$  is a *period* of a recurring decimal if starting from some place the digits of the decimal repeat every  $k$  digits. For instance,  $0.12345345345\cdots$  has smallest period 3 while  $0.142857142587142587\cdots$  has smallest period 6.)
6. Let  $O$  be the centre of the excircle, touching the side  $BC$  of  $\triangle ABC$ .  $M$  is the midpoint of  $AC$  and  $P$  is the intersection of lines  $MO$  and  $BC$ . Show that if  $\angle BAC = 2\angle ACB$ , then  $AB = BP$ .

### Test 2

1. Find all functions  $f: \mathbb{R} \rightarrow \mathbb{R}$  such that

- (i) the set  $\left\{ \frac{f(x)}{x} \mid x \neq 0 \right\}$  is finite,
- (ii)  $f(3x - 1 - f(x)) = 3(f(x) - 1 - 3x)$  for all  $x \in \mathbb{R}$ .

2. Let  $a, b, c$  be the sides of a triangle, and  $T$  its area. Prove that

$$a^2 + b^2 + c^2 \geq 4\sqrt{3}T + (a-b)^2 + (b-c)^2 + (c-a)^2.$$

When does the equality hold?



3. Given 2009 points in the plane so that no three points lie on a line, show that there exist at least 1344021 triangles, each of which is formed by any three of the 2009 points, and each of the triangles does not contain any of the remaining 2006 points.
4. How many odd coefficients are there in the expansion of  $(x^2 - x + 1)^{2009}$ ?

## Year 2011

### Test 1

1. Let  $I$  be the incentre of a triangle  $ABC$  with  $AB > AC$ . A line through  $I$  perpendicular to  $AI$  cuts  $BC$  produced at  $P$ . Show that  $IP^2 = BP \cdot CP$ .
2. Given that  $a$  and  $b$  are natural numbers, when  $a^2 + b^2$  is divided by  $a + b$ , the quotient is  $Q$  and the remainder is  $R$ , where  $R < a + b$ . Find all possible pairs of numbers  $a$  and  $b$  such that  $Q^2 + R = 2010$ .
3. Let  $a, b, c$  and  $d$  be positive real numbers, all larger than or equal to 1. Prove that

$$(i) \quad abcd(a + b + c + d - 4)^4 \geq (a - 1)(b - 1)(c - 1)(d - 1)(a + b + c + d)^4,$$

$$(ii) \quad \frac{d}{a + b + c} + \frac{a}{b + c + d} + \frac{b}{c + d + a} + \frac{c}{d + a + b} \geq \frac{4}{3}.$$

4. Let  $ABCD$  be a convex quadrilateral.  $E$  and  $F$  are points on the diagonal  $AC$  such that  $E$  and  $F$  are interior points of triangles  $ABD$  and  $BCD$  respectively. Suppose  $BE$  extend cuts  $AD$  at  $P$ ,  $DE$  extend cuts  $AB$  at  $Q$ ,  $DF$  extend cuts  $BC$  at  $R$  and  $BF$  extend cuts  $DC$  at  $S$ . Show that the three lines  $QP, BD$  and  $RS$  are either parallel or concurrent.

5. Evaluate  $\frac{\sqrt{45 + \sqrt{1}} + \sqrt{45 + \sqrt{2}} + \dots + \sqrt{45 + \sqrt{2024}}}{\sqrt{45 - \sqrt{1}} + \sqrt{45 - \sqrt{2}} + \dots + \sqrt{45 - \sqrt{2024}}}.$

6. In a city the phone numbers should consist of exactly eight digits, and 0 is not allowed as a digit in them, (i.e., only  $1, 2, 3, \dots, 9$  may be used). It is required that every two phone numbers to either be different in at least two places, or have digits separated by at least two units, in at least one of the eight places. (For examples, both 12345678 and 23345678 may be selected if there is no more conflict with other numbers, or both 12345678 and 92345678 may be selected, again if there is no more conflict with other numbers, but at most one of 12345678 and 12345679 may be selected.)

What is the maximum number of phone numbers that can be selected, satisfying the requirements? In how many ways can one select this amount of phone numbers?

### Test 2

1. Given a triangle  $ABC$ , let  $M$  be the midpoint of  $BC$ . The circle passing through  $A$  and tangent to  $BC$  at  $M$  cuts  $AB$  and  $AC$  at  $D$  and  $E$  respectively. Suppose  $B, C, E$  and  $D$  are concyclic. Show that  $AB = AC$ .
2. Suppose  $a_1 = 4$  and  $a_{n+1} \geq a_n + n + 2$  for all natural numbers  $n$ . Show that

$$\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_{2010}} < \frac{3}{2}.$$

3. Among a group of 451 persons, every pair of persons together know exactly one other person (knowing is mutual). Denote by  $n$  the difference between the number of persons known by a member of the group who knows the largest number of persons among the group, and the number of persons known by another member who knows the smallest number of persons among the group. Find the maximum possible value of  $n$  and all other possible value(s) of  $n$ .
4. Let  $p$  be a prime.
- (a) Show that there exists a prime divisor  $q$  of  $\frac{p^p - 1}{p - 1}$  such that  $q \not\equiv 1 \pmod{p^2}$ .
- (b) Suppose  $q$  is a prime divisor of  $\frac{p^p - 1}{p - 1}$  satisfying  $q \not\equiv 1 \pmod{p^2}$ . Show that  $n^p - p \not\equiv 0 \pmod{q}$  for all positive integers  $n$ .

## Year 2012

### Test 1

1. Find all reals  $x, y$  and  $z$  such that

$$\begin{cases} (x+1)yz = 27, \\ (y+1)zx = 12, \\ (z+1)xy = 12. \end{cases}$$

2. Find a function  $f : \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}^{\geq 0}$  satisfying  $f(2x+1) = 4f(x) + 9$  for all  $x \geq 0$ . ( $\mathbb{R}^{\geq 0}$  is the set of nonnegative real numbers.)
3. In  $\triangle ABC$ ,  $\angle A = 30^\circ$  and  $AB = 4$ .  $D$  is the midpoint of  $AB$ . Determine the maximum possible value of  $\tan \angle DCB$ .
4. Among 45 persons in a party, it is found that every person knows at least one other person in the party (i.e. they are friends), and any two persons with the same number of friends in the party do not know each other. What is the maximum number of pairs of persons who are friends?
5.  $E$  and  $F$  are points on side  $AB$  of  $\triangle ABC$  such that  $AE = EF = FB$ .  $D$  is a point on the line  $BC$  such that  $BC$  is perpendicular to  $ED$ ,  $AD$  is perpendicular to  $CF$ . Suppose  $\angle CFA = 3\angle BDF$ . Determine the value of  $\frac{DB}{DC}$ .
6. Determine the number of positive integers  $m$  for which there exist nonnegative integers  $x_0, x_1, x_2, \dots, x_{2011}$  such that  $m^{x_0} = m^{x_1} + m^{x_2} + \dots + m^{x_{2011}}$ .

### Test 2

1. Let  $x, y, z, u, v$  and  $w$  be integers satisfying  $x^2 + y^2 = u^2$ ,  $x^2 + z^2 = v^2$  and  $y^2 + z^2 = w^2$ . Find an integer  $\ell$  so that  $518000 < \ell < 518518$  and  $\ell$  divides  $xyzuvw$ .
2. Let  $ABC$  be a scalene triangle (with no two sides equal). A circle touching sides  $AB$  and  $BC$  intersects the median  $AM$  at  $P$  and  $Q$ . Another circle touching sides  $AC$  and  $BC$  also intersects  $AM$  at  $P$  and  $Q$ . Prove that the two circles are the same circle.

3. Let  $n$  be a positive integer larger than or equal to 2. On an  $n \times n$  chessboard,  $r$  chess pieces have been placed such that

- (i) there is at least one chess piece in each row;
- (ii) there is at least one chess piece in each column; and
- (iii) there is at least one chess piece in each diagonal (here a ‘diagonal’ refers to the collection of cells whose centres lie on a straight line which makes an angle of  $45^\circ$  with a side of the chessboard). (For instance, in the  $3 \times 3$  chessboard, the  $1 \times 1$  cell forms a diagonal; the  $1 \times 2$  cell and the  $2 \times 1$  cell together form a diagonal; the  $1 \times 1$  cell,  $2 \times 2$  cell and the  $3 \times 3$  cell also form a diagonal (a main diagonal), etc.)

Find, in terms of  $n$ , the smallest possible value of  $r$ .

4. Let  $f(x)$  be a continuous function defined on the set of real numbers such that for any real numbers  $x$ ,

- (i)  $f(x+1) = f(x) + 1$ , and
- (ii)  $f(x^3) = (f(x))^3$ .

Find all possible  $f(x)$ .

## Year 2013

### Test 1

1.  $a$  and  $b$  are natural numbers such that the last digit of  $a^2 - ab + b^2$  is 0. Find the values of all possible second-to-last digits of  $a^2 - ab + b^2$ .
2. For any positive integer  $n$ , define  $f(n)$  as the sum of the squares of the digits of  $n$  (in base ten). For example,  $f(2012) = 2^2 + 0^2 + 1^2 + 2^2 = 9$ . Find the greatest integer  $n$  such that there exists a positive integer  $d$  with  $\underbrace{f(f(\dots f(n)\dots))}_{d \text{ times}} = n$ .
3. Let  $a, b$  be real numbers. It is given that the equation  $x^4 + ax^3 + bx^2 + ax + 1 = 0$  has at least one real root. Find the minimum value of  $a^2 + b^2$ .
4.  $S = \{(x, y) : x, y \text{ are integers, } 0 \leq x \leq 5 \text{ and } 0 \leq y \leq 5\}$  is a set of lattice points on the rectangular coordinate plane. How many triangles can be formed by taking any three elements of  $S$ ?
5. In a competition, each participant can get an integral score lying between 0 and 10 inclusive. It is found that the mean, median and mode of the scores are 3, 3, 7 respectively. Find the minimum number of participants.
6. Given a line  $\ell$  and four points  $P, Q, R$  and  $S$  (in that order) on the line, describe a straight edge (unmarked ruler) and compass construction producing a square  $ABCD$  such that  $P$  lies on the line  $AD$ ,  $Q$  on the line  $BC$ ,  $R$  on the line  $AB$  and  $S$  on the line  $CD$ .

### Test 2

1. Find a positive integer  $m$  such that  $x^3 \equiv 1 \pmod{m}$  has more than 2012 solutions. (Note: If  $x \equiv y \pmod{m}$ , then  $x$  and  $y$  are considered as the same solution.)
2. Let  $S = \{1, 2, 3, \dots, 2n\}$ , where  $n$  is a positive integer greater than or equal to 1. For any subset  $T$  of  $S$ ,  $T$  is called a *good* subset if in  $T$  the number of even elements is greater than the number of odd elements.
  - (i) Find the total number of good subsets of  $S$ .
  - (ii) Find the sum of all the elements in all the good subsets of  $S$ .
3. For any positive real numbers  $a, b, c$  satisfying  $a + b + c = 1$ , prove that
$$\sqrt{a^2 - 2bc + 2c} + \sqrt{b^2 - 2ca + 2a} + \sqrt{c^2 - 2ab + 2b} \geq \sqrt{5}.$$
4. Let  $D, E$  and  $F$  be the contact points of the incircle of  $\triangle ABC$  with sides  $BC, CA$  and  $AB$  respectively. Let  $I$  and  $I'$  be the incentres of  $\triangle ABC$  and  $\triangle DEF$ , respectively. Let  $\ell_A$  be the line passing through  $D$  and is parallel to  $AI'$ ,  $\ell_B$  be the line passing through  $E$  and is parallel to  $BI'$ , and  $\ell_C$  be the line passing through  $F$  and is parallel to  $CI'$ . Show that the lines  $\ell_A, \ell_B$  and  $\ell_C$  are concurrent.

## Year 2014

### Test 1

1. Let  $f$  be a function defined on nonzero real numbers such that for any nonzero  $x$ ,  $\frac{2}{x}f(-x) + f\left(\frac{1}{x}\right) = x$ . Find the value of  $f(8)$ .
2. Let  $ABC$  be an equilateral triangle.  $P$  is a point inside the triangle such that  $\angle CBP = 12^\circ$  and  $\angle ACP = 54^\circ$ . Find  $\angle CAP$ .
3. Given a triangle  $ABC$  with  $AB = BC = 1$  and  $CA = \sqrt{2}$ ,  $P$  is a point inside the triangle  $ABC$  such that  $\angle PAB = \angle PBC = \angle PCA$ . Find  $BP$ .
4. A number is *good* if the sum of its digits is 18 (for example 130518 is a good number). Find the number of good 6-digit numbers.
5. Find the largest integer  $n$  satisfying the following conditions:
  - (i)  $n^2$  can be expressed as the difference of two consecutive cubes;
  - (ii)  $2n + 79$  is a perfect square.
6. Find the sum of all positive integers  $m$  such that for each such  $m$ , there exists a positive integer  $n$  satisfying  $\left\lfloor \frac{(2m+1)^n}{m(3m+1)} \right\rfloor = 2013$ , where  $[x]$  denotes the greatest integer not exceeding  $x$ .

### Test 2

1. Let  $a = BC$ ,  $b = CA$  and  $c = AB$  be respectively the lengths of a triangle  $ABC$ ;  $i_a, i_b, i_c$  be respectively the lengths of the angle bisectors from  $A, B$  and  $C$ . Let  $R$  be the circumradius of the triangle. Prove that
$$ai_a + bi_b + ci_c < 9R^2.$$
2. Let  $ABC$  be an acute triangle and  $H$  be the perpendicular foot at the side  $BC$  from  $A$ . Let  $D$  and  $E$  be points on the segments  $AB$  and  $AC$ ,  $F$  and  $G$  be the perpendicular feet at  $BC$  from  $D$  and  $E$  respectively. Let  $P$  be the perpendicular foot at the segment  $DH$  from  $E$ . Suppose that  $DG, EF, AH$  are concurrent. Show that  $\angle APE = \angle CPE$ .
3. Let  $m, n$  be distinct positive integers not exceeding 2013 and  $d$  be their greatest common divisor. Suppose  $d^2$  divides  $3(m - n)$ . Find the greatest possible value of  $d(m + n)$ .
4. In a school there are 2013 boys and 2013 girls. For each pair of a boy and a girl, together they have to choose one (and only one) of 25 different clubs to join. Determine the maximum possible value of the integer  $k$ , such that no matter what the choices of the students are, there is a club with  $k$  or more members. (A boy and a girl in a club together are counted as two members.)

### Test 3

1. Let  $ABCDEF$  be a hexagon whose diagonals  $AD$ ,  $BE$  and  $CF$  intersect at one point, which is the midpoint of each diagonal. Prove that the area of the given hexagon is twice the area of the triangle  $ACE$ .
2. Find the greatest positive integer  $k$  for which the following holds: For all positive real numbers  $a, b$  and  $c$  with  $abc = 1$ , we have the inequality

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{k}{a+b+c+1} \geq \frac{k}{4} + 3.$$

3. A real number is put inside each cell of an  $n \times n$  table. Each time we may add a real number  $x$  to each of the cells of a single row or a single column, where the real number  $x$  may vary at each time. Find the maximum  $k$  for which it is always possible to make  $k$  of the cells zeros simultaneously after a finite number of steps.

### Test 4

1. A set  $M$  of real numbers will be called *special* if it has the properties:
  - (i) for each  $x, y \in M$ ,  $x \neq y$ , the numbers  $x+y$  and  $xy$  are not zeros and exactly one of them is rational.
  - (ii) for each  $x \in M$ ,  $x^2$  is irrational.

Find the maximum number of elements of a special set.

2. Let  $S$  be a proper subset of  $\mathbb{R}$  (i.e.  $S \neq \mathbb{R}$ ) having at least two elements. Suppose there exists a function  $f : \mathbb{R} \rightarrow \mathbb{R}$  satisfying the following conditions:
  - (i)  $f(a+x+y) + f(f(a)) + f(x) + f(y) = x+y$ ; and
  - (ii)  $f(axy) + f(a) + f(x)f(y) = xy$

for any real numbers  $a \notin S$  and  $x, y \in S$ . Find all such function(s)  $f$ .

3.  $2n+1$  distinct points are chosen on a circle and each two of them are connected with a vector going in one of the two possible directions. Let  $R$  be the number of triangles with the vertices at the given points such that the sum of the vectors going along the sides of the triangle is equal to zero, (i.e. starting from any vertex, the vectors will go back to itself). Find the smallest and the biggest possible values of  $R$ .



## Year 2015

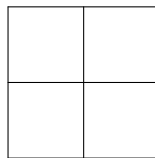
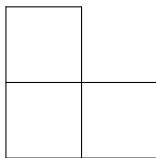
### Test 1

1. One day, one alien has come to the Earth. Every day, each alien does one of four things, each with a probability of one-fourth: (1) destroying himself, (2) splitting into two aliens, (3) splitting into three aliens, and (4) doing nothing. (A newly created alien also does one of these four things every day starting on the next day, but different aliens do not necessarily do the same thing and act independently.) Find the probability that there is eventually no alien on the Earth.
2. Find the maximum number of sets which simultaneously satisfy the following 3 conditions:
  - (i) any of the sets consists of 4 elements;
  - (ii) any two different sets have exactly 2 common elements;
  - (iii) no two elements are common to all the sets.
3. Let  $ABC$  be a triangle with  $\angle ABC > \angle BCA$  and  $\angle BCA \geq 30^\circ$ . The angle bisectors of  $\angle ABC$  and  $\angle BCA$  meet the opposite sides of the triangle at the points  $D$  and  $E$ , respectively. The line  $BD$  intersects the line  $CE$  at  $P$ . Assume that  $PD = PE$  and that the incircle of the triangle  $ABC$  has radius 1. Determine the largest possible length of  $BC$ .
4. Let  $a_n = 6 \underbrace{77 \dots 7}_n$ . Is it possible to find infinitely many multiples of  $a_{2014}$  in the sequence  $\{a_n\}$ ?
5. There are 100 squares in a  $10 \times 10$  grid table. Each square is painted in one colour. What is the maximum number of colours that may be used, so that in each row, and also in each column, the number of different colours used does not exceed five?
6.  $ABC$  is a triangle with integral sides.  $M$  is the midpoint of  $BC$ . The incircle with centre  $I$  touches  $AB$  and  $AC$  at  $E$  and  $F$  respectively and  $D$  is the projection of  $M$  on  $EF$ . Suppose that  $ADMI$  is a parallelogram and  $AB + BC + CA = 65$ . Find  $AB \times BC \times CA$ .

### Test 2

1. Assume the dimensions of an answer sheet to be 297 mm by 210 mm. Suppose that your pen leaks and makes some non-intersecting ink stains on the answer sheet. It turns out that the area of each ink stain does not exceed  $1 \text{ mm}^2$ . Moreover, any line parallel to an edge of the answer sheet intersects at most one ink stain. Prove that the total area of the ink stains is at most  $253.5 \text{ mm}^2$ . (You may assume a stain is a connected piece.)

2. Let  $\{a_n\}$  be a sequence of positive integers. It is given that  $a_1 = 1$ , and for  $n \geq 1$ ,  $a_{n+1}$  is the smallest positive integer greater than  $a_n$  which satisfies the following condition: for any integers  $i, j, k$ , with  $1 \leq i, j, k \leq n+1$ ,  $a_i + a_j \neq 3a_k$ . Find  $a_{2015}$ .
3. Let  $ABC$  be an equilateral triangle, and let  $D$  be a point on  $AB$  between  $A$  and  $B$ . Next, let  $E$  be a point on  $AC$  with  $DE$  parallel to  $BC$ . Further, let  $F$  be the midpoint of  $CD$  and  $G$  the circumcentre of triangle  $ADE$ . Determine the interior angles of triangle  $BFG$ .
4. A  $11 \times 11$  grid is to be covered completely without overlapping by some  $2 \times 2$  squares and  $L$ -shapes each composed of three unit cells. Determine the smallest number of  $L$ -shapes used. (Each shape must cover some grids entirely and cannot be placed outside the  $11 \times 11$  grid. The  $L$ -shapes may be reflected or rotated when placed on the grid.)



### Test 3

1. Consider six points in the interior of a square of side length 3. Prove that among the six points, there are two whose distance is less than 2.
2. Circles  $\omega_1, \omega_2$  with equal radius  $R$  intersect at two points. The line joining centres of these circles intersects  $\omega_1$  at points  $A, C$  and intersects  $\omega_2$  at points  $B, D$ . ( $B$  is between  $A$  and  $C$ ,  $C$  is between  $B$  and  $D$ ). Draw a circle  $\omega_3$  with diameter  $AD$ , tangent internally with  $\omega_1$  at  $A$  and tangent internally with  $\omega_2$  at  $D$ . Let  $\ell_1$  be the tangent line to the circle  $\omega_1$  at point  $C$  and let  $\ell_2$  be the tangent line to the circle  $\omega_2$  at point  $B$ .  $\ell_2$  intersects  $\omega_3$  at  $K$ . Tangent lines from the point  $K$  to the circle  $\omega_1$  intersect  $\ell_1$  at  $P$  and  $Q$ . Find the length of the line segment  $PQ$ .
3. Determine the maximum possible value of real number  $k$ , for which the inequality

$$\frac{a}{1 + 9bc + k(b - c)^2} + \frac{b}{1 + 9ca + k(c - a)^2} + \frac{c}{1 + 9ab + k(a - b)^2} \geq \frac{1}{2}$$

is satisfied for every choice of nonnegative real numbers  $a, b, c$  satisfying  $a + b + c = 1$ .

# Test 4

1. Let  $\ell(n)$  denote the greatest odd factor of any natural number  $n$ . Find the sum

$$\ell(1) + \ell(2) + \ell(3) + \cdots + \ell(2^{2015}).$$

2. Given triangle  $ABC$ , let  $AL$  be the angle bisector of  $\angle BAC$  with  $L$  on  $BC$ . Let the incircle of  $\triangle ABC$  touch the sides  $AB$  and  $BC$  at points  $P$  and  $Q$  respectively. Let  $X$  be the intersection point of the lines  $AQ$  and  $LP$ . Show that the lines  $BX$  and  $AL$  are perpendicular.
3. Determine all  $n \times n$  tables of nonnegative integers with rows and columns labelled by  $0, 1, 2, \dots, n-1$  such that for each  $0 \leq i, j \leq n-1$ , the number in the cell  $(i, j)$  is the number of  $i$ 's of row  $j$ . One such table can be seen below.

|       | column 0 | column 1 | column 2 | column 3 | column 4 |
|-------|----------|----------|----------|----------|----------|
| row 0 | 1        | 0        | 3        | 3        | 4        |
| row 1 | 1        | 3        | 2        | 1        | 1        |
| row 2 | 0        | 1        | 0        | 1        | 0        |
| row 3 | 2        | 1        | 0        | 0        | 0        |
| row 4 | 1        | 0        | 0        | 0        | 0        |

(The value in the cell  $(2, 3)$  is 1, for there is one “2” in row 3; the value in the cell  $(4, 2)$  is 0, for there is no “4” in row 2, etc.)

## Year 2016

### Test 1

1. Find all natural numbers  $n$  such that  $n$ ,  $n^2 + 10$ ,  $n^2 - 2$ ,  $n^3 + 6$  and  $n^5 + 36$  are all prime numbers.
2. Find the largest possible positive integer  $n$ , so that there exist  $n$  distinct positive real numbers  $x_1, x_2, \dots, x_n$  satisfying the following inequality: for any  $1 \leq i, j \leq n$ ,

$$(3x_i - x_j)(x_i - 3x_j) \geq (1 - x_i x_j)^2.$$

3. Let  $\triangle ABC$  be a triangle such that  $AB \neq AC$ . The incircle with centre  $I$  touches  $BC$  at  $D$ . Line  $AI$  intersects the circumcircle  $\Gamma$  of  $\triangle ABC$  at  $M$ , and  $DM$  again meets  $\Gamma$  at  $P$ . Find  $\angle API$ .
4. Find all triples of integers  $(m, p, q)$  such that  $2^m p^2 + 1 = q^7$ , where  $m > 0$ , and  $p$  and  $q$  are prime numbers, or show that they do not exist.
5. Let  $ABCD$  be inscribed in a circle with centre  $O$ . Let  $E$  be the intersection point of  $AC$  and  $BD$ .  $M$  and  $N$  are the midpoints of the arcs  $AB$  and  $CD$  respectively (the arcs not containing any other vertices). Let  $P$  be the intersection point of  $EO$  and  $MN$ . Suppose  $BC = 5$ ,  $AC = 11$ ,  $BD = 12$  and  $AD = 10$ . Find  $\frac{MN}{NP}$ .
6. 4031 lines are drawn on the plane. No two lines are parallel or perpendicular, and no three lines meet at one point. Determine the maximum number of acute-angled triangles that may be formed.

### Test 2

1. During a school year 44 competitions were held. Exactly 7 students won in each of the competitions. For any two competitions, there exists exactly 1 student who won in both competitions. Is it true that there exists a student who won all of the competitions?
2. Let  $\Gamma$  be a circle and  $AB$  be a diameter. Let  $\ell$  be a line outside the circle, and is perpendicular to  $AB$ . Let  $X, Y$  be two points on  $\ell$ . If  $X'$  and  $Y'$  are two points on  $\ell$  such that  $AX$  and  $BX'$  intersect on  $\Gamma$  and such that  $AY$  and  $BY'$  intersect on  $\Gamma$ , prove that the circumcircles of the triangles  $AXY$  and  $AX'Y'$  intersect at a point on  $\Gamma$  other than  $A$ , or the three circles are tangent at  $A$ .
3. Let  $a, b$  and  $c$  be positive real numbers satisfying  $abc = 1$ . Determine the smallest possible value of  $\frac{a^3 + 8}{a^3(b + c)} + \frac{b^3 + 8}{b^3(c + a)} + \frac{c^3 + 8}{c^3(a + b)}$ .

4. Mable and Nora play a game according to the following steps in order.
  - (1) Mable writes down any 2015 distinct prime numbers in ascending order in a row. The product of these primes is Mable's score.
  - (2) Nora writes down a positive integer.
  - (3) Mable draws a vertical line between two adjacent primes she has written in step (1), and compute the product of the prime(s) on the left of the vertical line.
  - (4) Nora must add the product obtained by Mable in step (3) to the number she has written in step (2), and the sum becomes Nora's score.

If Mable's and Nora's scores have a common factor greater than 1, Mable wins. Otherwise Nora wins.

Who has a winning strategy?

### Test 3

1. Find all prime numbers  $p$  and  $q$  such that  $p^2 \mid q^3 + 1$  and  $q^2 \mid p^6 - 1$ .
2. Suppose that  $I$  is the incentre of triangle  $ABC$ . The perpendicular to line  $AI$  from point  $I$  intersects sides  $AC$  and  $AB$  at points  $B'$  and  $C'$  respectively. Points  $B_1$  and  $C_1$  are placed on half-lines  $BC$  and  $CB$  respectively, in such a way that  $AB = BB_1$  and  $AC = CC_1$ . If  $T$  is the second intersection point of the circumcircles of triangles  $AB_1C'$  and  $AC_1B'$  prove that the circumcentre of triangle  $ATI$  lies on the line  $BC$ .
3. 2016 circles with radius 1 are lying on the plane. Among these 2016 circles, show that one can select a collection  $C$  of 27 circles satisfying the following: either every pair of two circles in  $C$  intersects or every pair of two circles in  $C$  does not intersect.

### Test 4

1. Let  $O$  be the circumcentre of a triangle  $ABC$ , and let  $\ell$  be the line going through the midpoint of the side  $BC$  and is perpendicular to the bisector of  $\angle BAC$ . Determine the value of  $\angle BAC$  if the line  $\ell$  goes through the midpoint of the line segment  $AO$ .
2. Determine all positive integers  $n$  for which there exist pairwise distinct positive real numbers  $a_1, a_2, \dots, a_n$  satisfying  $\left\{ a_i + \frac{(-1)^i}{a_i} \mid 1 \leq i \leq n \right\} = \{ a_i \mid 1 \leq i \leq n \}$ .
3. Let  $p$  be a prime number greater than 5. Suppose there is an integer  $k$  satisfying that  $k^2 + 5$  is divisible by  $p$ . Prove that there are positive integers  $m$  and  $n$  such that  $p^2 = m^2 + 5n^2$ .

## Year 2017

### Test 1

1. In  $\triangle ABC$ , let  $AD$  be the angle bisector of  $\angle BAC$ , with  $D$  on  $BC$ . The perpendicular from  $B$  to  $AD$  intersects the circumcircle of  $\triangle ABD$  at  $B$  and  $E$ . Prove that  $E, A$  and the circumcentre  $O$  of  $\triangle ABC$  are collinear.
2. In a committee there are  $n$  members. Each pair of members are either friends or enemies. Each committee member has exactly three enemies. It is also known that for each committee member, an enemy of his friend is automatically his own enemy. Find all possible value(s) of  $n$ .
3. Let  $f(x)$  be a monic cubic polynomial with  $f(0) = -64$  and all roots of  $f(x)$  are nonnegative real numbers. What is the largest possible value of  $f(-1)$ ? (A polynomial is *monic* if its leading coefficient is 1.)
4. Consider the sequences with 2016 terms formed by the digits 1, 2, 3 and 4. Find the number of those sequences containing an even number of the digit 1.
5. Find the first digit after the decimal point of the number  $\frac{1}{1009} + \frac{1}{1010} + \cdots + \frac{1}{2016}$ .
6. Let  $a_1, a_2, a_3, \dots$  and  $b_1, b_2, b_3, \dots$  be infinite sequences of real numbers satisfying  $a_{n+1} + b_{n+1} = \frac{a_n + b_n}{2}$  and  $a_{n+1}b_{n+1} = \sqrt{a_nb_n}$  for all  $n \geq 1$ . Suppose  $b_{2016} = 1$  and  $a_1 > 0$ . Find all possible value(s) of  $a_1$ .

### Test 2

1. Let  $\{a_n\}$  be a sequence of integers satisfying the following condition for all positive integral values of  $n$ :  $a_n + a_{n+1} = 2a_{n+2}a_{n+3} + 2016$ . Find all possible values of  $a_1$  and  $a_2$ .
2. Let  $ABCDEF$  be a convex hexagon such that  $\angle ACE = \angle BDF$  and  $\angle BCA = \angle EDF$ . Let  $A_1 = AC \cap FB$ ,  $B_1 = BD \cap AC$ ,  $C_1 = CE \cap BD$ ,  $D_1 = DF \cap CE$ ,  $E_1 = EA \cap DF$  and  $F_1 = FB \cap EA$ . Suppose  $B_1, C_1, D_1, F_1$  lie on the same circle  $\Gamma$ . The circumcircles of  $\triangle BB_1F_1$  and  $\triangle ED_1F_1$  meet at  $F_1$  and  $P$ . The line  $F_1P$  meets  $\Gamma$  again at  $Q$ . Prove that  $B_1D_1$  and  $QC_1$  are parallel. (Here, we use  $\ell_1 \cap \ell_2$  to denote the intersection point of lines  $\ell_1$  and  $\ell_2$ .)

3. Let  $G$  be a simple graph with  $n$  vertices and  $m$  edges. Two vertices are called *neighbours* if there is an edge between them. It turns out that  $G$  does not contain any cycle of length from 3 to  $2k$  (inclusive), where  $k \geq 2$  is a given positive integer.
  - (a) Prove that it is possible to pick a nonempty set  $S$  of vertices of  $G$  such that every vertex in  $S$  has at least  $\left\lceil \frac{m}{n} \right\rceil$  neighbours that are in  $S$ . ( $\lceil x \rceil$  stands for the smallest integer larger than or equal to  $x$ .)
  - (b) Suppose a set  $S$  as described in (a) is chosen. Let  $H$  be the graph consisting of the vertices in  $S$  and the edges between those vertices only. Let  $v$  be a vertex of  $H$ . Prove that at least  $\left\lceil \left( \frac{m}{n} - 1 \right)^k \right\rceil$  vertices of  $H$  can be reached by starting at  $v$  and travelling along the edges of  $H$  for at most  $k$  steps. (Note that  $v$  itself is a vertex that satisfies this condition, since it can be reached by starting at  $v$  and travelling along the edges of  $H$  for 0 steps.)
4. Let  $n$  be a positive integer with the following property:  $2^n - 1$  divides a number of the form  $m^2 + 81$ , where  $m$  is a positive integer. Find all possible  $n$ .

### Test 3

1. (a) Does there exist 5 circles in the plane such that each circle passes through exactly 3 centres of other circles?  
 (b) Does there exist 6 circles in the plane such that each circle passes through exactly 3 centres of other circles?
2. Suppose all of the 200 integers lying in between (and including) 1 and 200 are written on a blackboard. Suppose we choose exactly 100 of these numbers and circle each one of them. By the *score* of such a choice, we mean the square of the difference between the sum of the circled numbers and the sum of the non-circled numbers. What is the average of the scores over all possible choices for 100 numbers?
3. At a mathematical competition  $n$  students work on 6 problems each one with three possible answers. After the competition, the Jury found that for every two students the number of the problems for which these students have the same answers is 0 or 2. Find the maximum possible value of  $n$ .

### Test 4

1. Decide if there is a permutation  $a_1, a_2, \dots, a_{6666}$  of the numbers  $1, 2, \dots, 6666$  with the property that the sum  $k + a_k$  is a perfect square for all  $k = 1, 2, \dots, 6666$ .

2. Two circles  $\omega_1$  and  $\omega_2$  centered at  $O_1$  and  $O_2$ , respectively, meet at points  $A$  and  $B$ . A line through  $B$  meets  $\omega_1$  again at  $C$  and  $\omega_2$  again at  $D$ . The tangents to  $\omega_1$  and  $\omega_2$  at  $C$  and  $D$ , respectively, meet at  $E$ , and the line  $AE$  meets the circle  $\omega$  through  $A, O_1, O_2$  again at  $F$ . Prove that the length of the segment  $EF$  is equal to the diameter of  $\omega$ .
3. Let a sequence of real numbers  $a_0, a_1, a_2, \dots$  satisfies the condition

$$\sum_{n=0}^m a_n \cdot (-1)^n \cdot \binom{m}{n} = 0$$

for all sufficiently large values of  $m$ . Show that there exists a polynomial  $P$  such that  $a_n = P(n)$  for all  $n \geq 0$ .



## Year 2018

### Test 1

1. Find all positive integer(s)  $n$  such that  $n^2 + 32n + 8$  is a perfect square.
2. Find all polynomials  $f$  such that  $f$  has nonnegative integer coefficients,  $f(1) = 7$  and  $f(2) = 2017$ .
3. In a school there are 1200 students. Each student must join exactly  $k$  clubs. Given that there is a common club joined by every 23 students, but there is no common club joined by all 1200 students, find the smallest possible value of  $k$ .
4. Find infinitely many positive integers  $m$  such that for each such  $m$ , the number  $\frac{2^{m-1} - 1}{8191m}$  is an integer.
5. In a group of 2017 persons, any pair of persons has exactly one common friend (other than the pair of persons). Determine the smallest possible value of the difference between the numbers of friends of the person with the most friends and the person with the least friends in such a group.
6. A triangle  $ABC$  has its orthocentre  $H$  distinct from its vertices and from the circumcentre  $O$  of  $\triangle ABC$ . Denote by  $M, N$  and  $P$  respectively the circumcentres of triangles  $HBC$ ,  $HCA$  and  $HAB$ . Show that the lines  $AM, BN, CP$  and  $OH$  are concurrent.

### Test 2

1. Let  $ABC$  be a triangle with  $AB = AC$ . A circle  $\Gamma$  lies outside triangle  $ABC$  and is tangent to line  $AC$  at  $C$ . Point  $D$  lies on  $\Gamma$  such that the circumcircle of triangle  $ABD$  is internally tangent to  $\Gamma$ . Segment  $AD$  meets  $\Gamma$  again at  $E$ . Prove that  $BE$  is tangent to  $\Gamma$ .
2. There are three piles of coins, with  $a, b$  and  $c$  coins respectively, where  $a, b, c \geq 2015$  are positive integers. The following operations are allowed:
  - (1) Choose a pile with an even number of coins and remove all coins from this pile. Add coins to each of the remaining two piles with amount equal to half of that removed; or
  - (2) Choose a pile with an odd number of coins and at least 2017 coins. Remove 2017 coins from this pile. Add 1009 coins to each of the remaining two piles.Suppose there are sufficiently many spare coins. Find all ordered triples  $(a, b, c)$  such that after some finite sequence of allowed operations, there exists a pile with at least  $2017^{2017}$  coins.
3. Find all functions  $f : \mathbb{R} \rightarrow \mathbb{R}$  such that

$$f(f(xy - x)) + f(x + y) = yf(x) + f(y)$$

for all real numbers  $x$  and  $y$ .

4. In triangle  $ABC$  with incentre  $I$ , let  $M_A, M_B$  and  $M_C$  be the midpoints of  $BC, CA$  and  $AB$  respectively, and  $H_A, H_B$  and  $H_C$  be the feet of altitudes from  $A, B$  and  $C$  to the respective sides. Denote by  $\ell_b$  the line being tangent to the circumcircle of triangle  $ABC$  and passing through  $B$ , and denote by  $\ell'_b$  the reflection of  $\ell_b$  in  $BI$ . Let  $P_B$  be the intersection of  $M_A M_C$  and  $\ell_b$ , and let  $Q_B$  be the intersection of  $H_A H_C$  and  $\ell'_b$ . Define  $\ell_c, \ell'_c, P_C$  and  $Q_C$  analogously. If  $R$  is the intersection of  $P_B Q_B$  and  $P_C Q_C$ , prove that  $RB = RC$ .

### Test 3

- Does there exist a polynomial  $P(x)$  with integer coefficients such that  $P(1 + \sqrt[3]{2}) = 1 + \sqrt[3]{2}$  and  $P(1 + \sqrt{5}) = 2 + 3\sqrt{5}$ ?
- Given triangle  $ABC$ , let  $D$  be an inner point of the segment  $BC$ . Let  $P$  and  $Q$  be distinct inner points of the segment  $AD$ . Let  $K = BP \cap AC$ ,  $L = CP \cap AB$ ,  $E = BQ \cap AC$ ,  $F = CQ \cap AB$ . Given that  $KL \parallel EF$ , find all possible values of the ratio  $BD : DC$ .
- On a rectangular board with  $m$  rows and  $n$  columns, where  $m \leq n$ , some unit squares are coloured black in such a way that no two rows are alike. Find the biggest integer  $k$  such that every possible colouring to start with one can always colour  $k$  columns entirely red in such a way that no two rows are still alike.

### Test 4

- The altitudes  $AD$  and  $BE$  of acute triangle  $ABC$  intersect at  $H$ . Let  $F$  be the intersection of  $AB$  and a line that is parallel to the side  $BC$  and goes through the circumcentre of  $ABC$ . Let  $M$  be the midpoint of  $AH$ . Prove that  $\angle CMF = 90^\circ$ .
- For which natural number  $n$  is it possible to place natural numbers from 1 to  $3n$  on the edges of a right  $n$ -angled prism (on each edge there is exactly one number placed and each one is used exactly 1 time) in such a way that the sum of all the numbers that surround each face is the same?
- Find all primes  $p$  and all positive integers  $a$  and  $m$  such that  $a \leq 5p^2$  and  $(p-1)! + a = p^m$ .

## Year 2019

### Test 1

1. Let  $a$  be a real number. Suppose the function  $f(x) = \frac{a}{x-1} + \frac{1}{x-2} + \frac{1}{x-6}$  defined in the interval  $3 < x < 5$  attains its maximum at  $x = 4$ . Find the value of  $a$ .
2. A circle is circumscribed around an isosceles triangle whose two base angles are equal to  $x^\circ$ . Two points are chosen independently and randomly on the circle, and a chord is drawn between them. The probability that the chord intersects the triangle is  $\frac{14}{25}$ . Find the sum of the largest and smallest possible value of  $x$ .
3. Find an integral solution of the equation

$$\left\lfloor \frac{x}{1!} \right\rfloor + \left\lfloor \frac{x}{2!} \right\rfloor + \left\lfloor \frac{x}{3!} \right\rfloor + \cdots + \left\lfloor \frac{x}{10!} \right\rfloor = 2019.$$

(Note  $\lfloor u \rfloor$  stands for the greatest integer less than or equal to  $u$ .)

4. Let  $ABC$  be an acute-angled triangle such that  $\angle ACB = 45^\circ$ . Let  $G$  be the point of intersection of the three medians and let  $O$  be the circumcentre. Suppose  $OG = 1$  and  $OG \parallel BC$ . Determine the length of the segment  $BC$ .
5. Is it possible to choose 24 distinct points in the space such that no three of them lie on the same line, and choose 2019 distinct planes in a way that each plane passes through at least 3 of the chosen points and each triple of points belongs to one of the chosen planes?
6. If  $57a + 88b + 125c \geq 1148$ , where  $a, b, c > 0$ , what is the minimum value of

$$a^3 + b^3 + c^3 + 5a^2 + 5b^2 + 5c^2?$$

### Test 2

1. Determine all sequences  $p_1, p_2, p_3, \dots$  of prime numbers for which there exists an integer  $k$  such that the recurrence relation

$$p_{n+2} = p_{n+1} + p_n + k$$

holds for all positive integers  $n$ .

2. Let  $p$  be a prime number greater than 10. Prove that there exist positive integers  $m$  and  $n$  such that  $m + n < p$ , and  $5^m 7^n - 1$  is divisible by  $p$ .

3. Let  $\Gamma_1$  and  $\Gamma_2$  be two circles with different radii, with  $\Gamma_1$  the smaller one. The two circles meet at distinct points  $A$  and  $B$ .  $C$  and  $D$  are two points on the circles  $\Gamma_1$  and  $\Gamma_2$  respectively, and such that  $A$  is the midpoint of the segment  $CD$ .  $CB$  is extended to meet the circle  $\Gamma_2$  at  $F$ , while  $DB$  is extended to meet the circle  $\Gamma_1$  at  $E$ . The perpendicular bisector of  $CD$  and the perpendicular bisector of  $EF$  meet at  $P$ .
  - (a) Prove that  $\angle EPF = 2\angle CAE$ .
  - (b) Prove that  $AP^2 = CA^2 + PE^2$ .
4. We choose 100 points in the coordinate plane. Let  $N$  be the number of triples  $(A, B, C)$  of distinct chosen points such that  $A$  and  $B$  have the same  $y$ -coordinate, and  $B$  and  $C$  have the same  $x$ -coordinate. Find the greatest possible value that  $N$  can attain by considering all possible ways to choose the points.

### Test 3

1. Let  $\mathbb{Q}_{>0}$  denote the set of positive rational numbers. Determine all functions  $f : \mathbb{Q}_{>0} \rightarrow \mathbb{Q}_{>0}$  satisfying

$$f(x^2 f(y)^2) = f(x)^2 f(y)$$

for all  $x, y \in \mathbb{Q}_{>0}$ .

2. Let  $ABC$  be a triangle with  $AB = AC$ , and let  $M$  be the midpoint of  $BC$ . Let  $P$  be a point such that  $PB < PC$  and  $PA$  is parallel to  $BC$ . Let  $X$  and  $Y$  be points on the lines  $PB$  and  $PC$ , respectively, so that  $B$  lies on the segment  $PX$ ,  $C$  lies on the segment  $PY$ , and  $\angle PXM = \angle PYM$ . Prove that the quadrilateral  $APXY$  is cyclic.
3. Let  $n$  be a given positive integer. Sisyphus performs a sequence of turns on a board consisting of  $n + 1$  squares in a row, number 0 to  $n$  from left to right. Initially,  $n$  stones are put into square 0, and the other squares are empty. At every turn, Sisyphus chooses any nonempty square, say with  $k$  stones, takes one of those stones and moves it to the right by at most  $k$  squares (the stones should stay within the board). Sisyphus' aim is to move all  $n$  stones to square  $n$ .

Prove that Sisyphus cannot reach the aim in less than

$$\left\lceil \frac{n}{1} \right\rceil + \left\lceil \frac{n}{2} \right\rceil + \left\lceil \frac{n}{3} \right\rceil + \cdots + \left\lceil \frac{n}{n} \right\rceil$$

turns. (As usual,  $\lceil x \rceil$  stands for the least integer not smaller than  $x$ .)

#### Test 4

1. Determine all pairs  $(n, k)$  of distinct positive integers such that there exists a positive integer  $s$  for which the number of divisors of  $sn$  and of  $sk$  are equal.
2. Let  $n \geq 3$  be an integer. Prove that there exists a set of  $2n$  positive integers satisfying the following property: For every  $m = 2, 3, \dots, n$  the set  $S$  can be partitioned into two subsets with equal sums of elements, with one of the subsets of cardinality  $m$ .
3. Find the maximal value of

$$S = \sqrt[3]{\frac{a}{b+7}} + \sqrt[3]{\frac{b}{c+7}} + \sqrt[3]{\frac{c}{d+7}} + \sqrt[3]{\frac{d}{a+7}},$$

where  $a, b, c, d$  are nonnegative real numbers which satisfy  $a + b + c + d = 100$ .

## Year 2020

### Test 1

1. Find all functions  $f : \mathbb{N} \rightarrow \mathbb{N}$  such that for every positive integer  $n$  the following is valid: If  $d_1, d_2, \dots, d_s$  are all the positive divisors of  $n$ , then

$$f(d_1)f(d_2)\cdots f(d_s) = n.$$

2. Let  $D$  be an arbitrary point inside  $\triangle ABC$ . Let  $\Gamma$  be the circumcircle of  $\triangle BCD$ . The external angle bisector of  $\angle ABC$  meets  $\Gamma$  again at  $E$ . The external angle bisector of  $\angle ACB$  meets  $\Gamma$  again at  $F$ . The line  $EF$  meets the extension of  $AB$  and  $AC$  at  $P$  and  $Q$  respectively. Prove that the circumcircles of  $\triangle BFP$  and  $\triangle CEQ$  always pass through the same fixed point regardless of the position of  $D$ . (Assume all the labelled points are distinct.)
3. Given a list of integers  $2^1 + 1, 2^2 + 1, \dots, 2^{2019} + 1$ , Adam chooses two different integers from the list and computes their greatest common divisor. Find the sum of all possible values of this greatest common divisor.
4. Find the total number of primes  $p < 100$  such that  $\lfloor (2 + \sqrt{5})^p \rfloor - 2^{p+1}$  is divisible by  $p$ . Here  $\lfloor x \rfloor$  denotes the greatest integer less than or equal to  $x$ .
5. In  $\triangle ABC$ , let  $D$  be a point on side  $BC$ . Suppose the incircle  $\omega_1$  of  $\triangle ABD$  touches sides  $AB$  and  $AD$  at  $E, F$  respectively, and the incircle  $\omega_2$  of  $\triangle ACD$  touches sides  $AD$  and  $AC$  at  $F, G$  respectively. Suppose the segment  $EG$  intersects  $\omega_1$  and  $\omega_2$  again at  $P$  and  $Q$  respectively. Show that line  $AD$ , tangent of  $\omega_1$  at  $P$  and tangent of  $\omega_2$  at  $Q$  are concurrent.
6. For a sequence with some ones and zeros, we count the number of continuous runs of equal digits in it. (For example the sequence 011001010 has 7 continuous runs: 0, 11, 00, 1, 0, 1, 0.) Find the sum of the number of all continuous runs for all possible sequences with 2019 ones and 2019 zeros.

### Test 2

1. Let  $\triangle ABC$  be an acute triangle with incentre  $I$  and orthocentre  $H$ .  $AI$  meets the circumcircle of  $\triangle ABC$  again at  $M$ . Suppose the length  $IM$  is exactly the circumradius of  $\triangle ABC$ . Show that  $AH \geq AI$ .
2. Suppose there are 2019 distinct points in a plane and the distances between pairs of them attain  $k$  different values. Prove that  $k$  is at least 44.
3. Two circles  $\Gamma$  and  $\Omega$  intersect at two distinct points  $A$  and  $B$ . Let  $P$  be a point on  $\Gamma$ . The tangent at  $P$  to  $\Gamma$  meets  $\Omega$  at the points  $C$  and  $D$ , where  $D$  lies between  $P$  and  $C$ , and  $ABCD$  is a convex quadrilateral. The lines  $CA$  and  $CB$  meet  $\Gamma$  again at  $E$  and  $F$  respectively. The lines  $DA$  and  $DB$  meet  $\Gamma$  again at  $S$  and  $T$  respectively. Suppose the points  $P, E, S, F, B, T, A$  lie on  $\Gamma$  in this order. Prove that  $PC, ET, SF$  are parallel.

4. Find all real-valued functions  $f$  defined on the set of real numbers such that

$$f(f(x) + y) + f(x + f(y)) = 2f(xf(y))$$

for any real numbers  $x$  and  $y$ .

### Test 3

1. Let  $ABC$  be a triangle. Circle  $\Gamma$  passes through  $A$ , meets segments  $AB$  and  $AC$  again at points  $D$  and  $E$  respectively, and intersects segment  $BC$  at  $F$  and  $G$  such that  $F$  lies between  $B$  and  $G$ . The tangent to circle  $BDF$  at  $F$  and the tangent to circle  $CEG$  at  $G$  meet at point  $T$ . Suppose that points  $A$  and  $T$  are distinct. Prove that line  $AT$  is parallel to  $BC$ .
2. Find all triples  $(a, b, c)$  of positive integers such that  $a^3 + b^3 + c^3 = (abc)^2$ .
3. Let  $u_1, u_2, \dots, u_{2019}$  be real numbers satisfying

$$u_1 + u_2 + \dots + u_{2019} = 0 \quad \text{and} \quad u_1^2 + u_2^2 + \dots + u_{2019}^2 = 1.$$

Let  $a = \min(u_1, u_2, \dots, u_{2019})$  and  $b = \max(u_1, u_2, \dots, u_{2019})$ . Prove that

$$ab \leq -\frac{1}{2019}.$$

### Test 4

1. You are given a set of  $n$  blocks, each weighing at least 1; their total weight is  $2n$ . Prove that for every real number  $r$  with  $0 \leq r \leq 2n - 2$  you can choose a subset of the blocks whose total weight is at least  $r$  but at most  $r + 2$ .
2. Let  $x_1, x_2, \dots, x_n$  be different real numbers. Prove that

$$\sum_{1 \leq i \leq n} \prod_{j \neq i} \frac{1 - x_i x_j}{x_i - x_j} = \begin{cases} 0, & \text{if } n \text{ is even;} \\ 1, & \text{if } n \text{ is odd.} \end{cases}$$

3. Let  $ABCDE$  be a convex pentagon with  $CD = DE$  and  $\angle EDC \neq 2\angle ADB$ . Suppose that a point  $P$  is located in the interior of the pentagon such that  $AP = AE$  and  $BP = BC$ . Prove that  $P$  lies on the diagonal  $CE$  if and only if  $[BCD] + [ADE] = [ABD] + [ABP]$ . (Here  $[XYZ]$  stands for the area of triangle  $XYZ$ .)

## Year 2021

### Test 1

1. Find all real triples  $(a, b, c)$  satisfying

$$(2^{2a} + 1)(2^{2b} + 2)(2^{2c} + 8) = 2^{a+b+c+5}.$$

2. In  $\triangle ABC$ ,  $AC = kAB$ , with  $k > 1$ . The internal angle bisector of  $\angle BAC$  meets  $BC$  at  $D$ . The circle with  $AC$  as diameter cuts the extension of  $AD$  at  $E$ . Express  $\frac{AD}{AE}$  in terms of  $k$ .
3. On the table there are 20 coins of weights  $1, 2, 3, \dots, 15, 37, 38, 39, 40$  and  $41$  grams. They all look alike but their colours are all distinct. Now Miss Adams knows the weight and colour of each coin, but Mr. Bean knows only the weights of the coins. There is also a balance on the table, and each comparison of weights of two groups of coins is called an operation. Miss Adams wants to tell Mr. Bean which coin is the 1 gram coin by performing some operations. What is the minimum number of operations she needs to perform?
4. Let  $n$  be a positive integer. Is it possible to express  $n^2 + 3n + 3$  into the form  $ab$  with  $a$  and  $b$  being positive integers, and such that the difference between  $a$  and  $b$  is smaller than  $2\sqrt{n+1}$ ?
5. Let  $ABCD$  be an isosceles trapezoid with bases  $BC$  and  $AD$ . Suppose  $\angle BDC = 10^\circ$  and  $\angle BDA = 70^\circ$ . Show that  $AD^2 = BC(AD + AB)$ .
6. There is an  $n \times n$  chessboard where  $n \geq 4$  is a positive even number. The cells of the chessboard are coloured black and white such that adjacent cells sharing a common side have different colours. Let  $A$  and  $B$  be two interior cells (which means cells not lying on an edge of the chessboard) of distinct colours. Prove that a chess piece can move from  $A$  to  $B$  by moving across adjacent cells such that every cell of the chessboard is passed through exactly once.

### Test 2

1. Let  $S$  be a set of 2020 distinct points in the plane. Let

$$M = \{P : P \text{ is the midpoint of } XY \text{ for some distinct points } X, Y \text{ in } S\}.$$

Find the least possible value of the number of points in  $M$ .

2. Let  $f(x)$  be a polynomial with rational coefficients, and let  $\alpha$  be a real number. If

$$\alpha^3 - 2019\alpha = (f(\alpha))^3 - 2019f(\alpha) = 2021,$$

prove that  $(f^n(\alpha))^3 - 2019f^n(\alpha) = 2021$  for any positive integer  $n$ .

(Here, we define  $f^n(x) = \underbrace{f(f(f \cdots f(x) \cdots))}_{n \text{ times}}.$ )



3. Let  $\triangle ABC$  be an acute triangle with circumcircle  $\Gamma$ , and let  $P$  be the midpoint of the minor arc  $BC$  of  $\Gamma$ . Let  $AP$  meet  $BC$  at  $D$ , and let  $M$  be the midpoint of  $AB$ . Also, let  $E$  be the point such that  $AE \perp AB$  and  $BE \perp MP$ . Prove that  $AE = DE$ .
4. Does there exist a nonzero polynomial  $P(x)$  with integer coefficients satisfying both of the following conditions?
  - $P(x)$  has no rational root;
  - For every positive integer  $n$ , there exists an integer  $m$  such that  $n$  divides  $P(m)$ .

### Test 3

1. Suppose that  $a, b, c, d$  are positive real numbers satisfying  $(a+c)(b+d) = ac+bd$ . Find the smallest possible value of

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}.$$

2. Let  $ABC$  be an isosceles triangle with  $BC = CA$ , and let  $D$  be a point inside side  $AB$  such that  $AD < DB$ . Let  $P$  and  $Q$  be two points inside sides  $BC$  and  $CA$ , respectively, such that  $\angle DPB = \angle DQA = 90^\circ$ . Let the perpendicular bisector of  $PQ$  meet line segment  $CQ$  at  $E$ , and let the circumcircles of triangles  $ABC$  and  $CPQ$  meet again at point  $F$ , different from  $C$ . Suppose that  $P, E, F$  are collinear. Prove that  $\angle ACB = 90^\circ$ .
3. For a positive integer  $n$ , let  $d(n)$  be the number of positive divisors of  $n$ , and let  $\varphi(n)$  be the number of positive integers not exceeding  $n$  which are coprime to  $n$ . Does there exist a constant  $C$  such that

$$\frac{\varphi(d(n))}{d(\varphi(n))} \leq C$$

for all  $n \geq 1$ ?

### Test 4

1. Let  $n$  be a positive integer. Find the number of permutations  $a_1, a_2, \dots, a_n$  of the sequence  $1, 2, \dots, n$  satisfying

$$a_1 \leq 2a_2 \leq 3a_3 \leq \dots \leq na_n.$$

2. Let  $\mathcal{A}$  denote the set of all polynomials in three variables  $x, y, z$  with integer coefficients. Let  $\mathcal{B}$  denote the subset of  $\mathcal{A}$  formed by all polynomials which can be expressed as

$$(x+y+z)P(x,y,z) + (xy+yz+zx)Q(x,y,z) + xyzR(x,y,z)$$

with  $P, Q, R \in \mathcal{A}$ . Find the smallest nonnegative integer  $n$  such that  $x^i y^j z^k \in \mathcal{B}$  for all nonnegative integers  $i, j, k$  satisfying  $i+j+k \geq n$ .

3. Let  $I$  and  $I_A$  be the incentre and the  $A$ -excentre of an acute-angled triangle  $ABC$  with  $AB < AC$ . Let the incircle meet  $BC$  at  $D$ . The line  $AD$  meets  $BI_A$  and  $CI_A$  at  $E$  and  $F$ , respectively. Prove that the circumcircles of triangles  $AID$  and  $I_AEF$  are tangent to each other.

## Year 2022

### Test 1

1. Let  $\{a_n\}$  be the sequence of rational numbers defined by  $a_0 = 2021$ , and  $a_{n+1} = a_n + \frac{2}{a_n}$ , for all  $n \geq 0$ . Can any  $a_n$ ,  $n \geq 1$ , be a square of a rational number?
2. Let  $a, b, c, d$  be roots of the equation  $x^4 + x + 1 = 0$ . Let  $a^5 + 2a + 1$ ,  $b^5 + 2b + 1$ ,  $c^5 + 2c + 1$ ,  $d^5 + 2d + 1$  be roots of the equation  $x^4 + px^3 + qx^2 + rx + s = 0$ . Find the value of  $p + 2q + 4r + 8s$ .
3. Four non-overlapping families dine at a restaurant together. Each family consists of one father, one mother and three children. There are five tables, which are red, blue, yellow, green and brown respectively. Every table can accommodate four people, and every person must sit at some table. How many ways are there to assign the 20 people to tables so that each child sits at the same table as at least one of his/her parents? (We do not distinguish between the four seats at the same table.)
4. Let  $ABCD$  be a cyclic quadrilateral with circumcentre  $O$ . Diagonals  $AC$  and  $BD$  meet at  $E$ .  $F$  and  $G$  are points on segments  $AB$  and  $CD$  respectively. Suppose  $AF = 15$ ,  $FB = 13$ ,  $BE = 30$ ,  $ED = 13$ ,  $DG = 7.5$  and  $GC = 6.5$ . Let  $P$  be a point such that  $PF \perp AB$  and  $PG \perp CD$ . Find  $\frac{PE}{PO}$ .
5. Let  $f(n) = \left| \binom{n}{0} + \binom{n}{3} + \binom{n}{6} + \cdots + \binom{n}{3\left[\frac{n}{3}\right]} - \frac{2^n}{3} \right|$ , where  $[x]$  is the greatest integer not exceeding  $x$ . Find  $f(1) + f(2) + \cdots + f(2021)$ .
6. There is a set of  $n$  01-sequences of length 200. Every pair of 01-sequences differ at least at 101 positions. (For example, the two 01-sequences of length 6, 111100 and 010001 differ at four positions, 1st, 3rd, 4th and 6th positions, counting from the left.) Is it possible that  $n \geq 101$ ?

### Test 2

1. Let  $ABC$  be a triangle. Let  $M$  be the midpoint of  $BC$ , and let  $G$  be the centroid of  $\triangle ABC$ . Let  $D$  be a point on the segment  $GM$ . A straight line passing through  $D$  meets the sides  $AB$  and  $AC$  at  $E$  and  $F$  respectively (with  $E, F \neq A$ ). Show that

$$[BEGF] + [CFGE] \geq \frac{4}{9}[ABC].$$

When does the equality hold? (Here  $[WXYZ]$  is the area of the polygon  $WXYZ$ , etc.)

2. Let  $P(x)$  be a polynomial with integer coefficients. Define a sequence  $\{a_n\}$  by  $a_0 = 0$  and  $a_n = P(a_{n-1})$  for all  $n \geq 1$ . Prove that if there exists a positive integer  $m$  for which  $a_m = 0$ , then  $a_1 = 0$  or  $a_2 = 0$ .
3. Let  $S$  be the set of all integers of the form  $x^2 + 3xy + 8y^2$  where  $x$  and  $y$  are integers.
  - (a) Show that if  $u$  and  $v$  are in  $S$ , then so is  $uv$ .
  - (b) Can an integer of the form  $23k + 7$ , with  $k$  an integer, belong to  $S$ ?
4. In a chess tournament there are 100 players. On each day of the tournament, each player is designated to be 'white', 'black' or 'idle', and each 'white' player will play a game against every 'black' player. (You may assume that all games fixed for the day can be finished within that day.) At the end of the tournament, it was found that any two players have met exactly once. What is the minimum duration of days that the tournament lasts?

### Test 3

1. Let  $n$  be an integer, and let  $A$  be a subset of  $\{0, 1, 2, 3, \dots, 5^n\}$  consisting of  $4n+2$  numbers. Prove that there exist  $a, b, c \in A$  such that  $a < b < c$  and  $c + 2a > 3b$ .
2. Find all positive integers  $n$  with the following property: the  $k$  positive divisors of  $n$  have a permutation  $(d_1, d_2, \dots, d_k)$  such that for every  $i = 1, 2, \dots, k$ , the number  $d_1 + d_2 + \dots + d_i$  is a perfect square.
3. Let  $ABCD$  be a quadrilateral inscribed in a circle  $\Omega$ . Let the tangent to  $\Omega$  at  $D$  intersect the rays  $BA$  and  $BC$  at points  $E$  and  $F$ , respectively. A point  $T$  is chosen inside the triangle  $ABC$  so that  $TE \parallel CD$  and  $TF \parallel AD$ . Let  $K \neq D$  be a point on the segment  $DF$  such that  $TD = TK$ . Prove that the lines  $AC, DT$  and  $BK$  intersect at one point.
4. Let  $n$  and  $k$  be two integers with  $n > k \geq 1$ . There are  $2n + 1$  students standing in a circle. Each student  $S$  has  $2k$  neighbours—namely, the  $k$  students closest to  $S$  on the right, and the  $k$  students closest to  $S$  on the left.  
 Suppose that  $n + 1$  of the students are girls, and the other  $n$  are boys. Prove that there is a girl with at least  $k$  girls among her neighbours.

## Year 2023

### Test 1

1. Mandy needs to wake up early for attending a mathematics contest. She has set an alarm in her smartphone every 15 minutes since 5:30 am. If an alarm is not pressed off by her or her mother (or anybody else), it will ring for a while, stop for a while, then will ring again 9 minutes later as the first ring, and so on (e.g. if the first alarm is not pressed off, it will ring again at 5:39 am). Also each alarm will work independently. Now suppose each ring-tone lasts for  $x$  minutes, and the smartphone has eventually rung for 50 minutes before Mandy wakes up at 6:30 am (assuming no one has pressed off any alarm before that). Find the value of  $x$ .
2. Let  $n$  be a positive integer. Show that if  $p$  is a prime dividing  $5^{4n} - 5^{3n} + 5^{2n} - 5^n + 1$ , then  $p \equiv 1 \pmod{4}$ .
3. A point  $P$  lies inside an equilateral triangle  $ABC$  such that  $AP = 15$  and  $BP = 8$ . Find the maximum possible value of the sum of areas of triangles  $ABP$  and  $BCP$ .
4. Let  $x, y, z$  be real numbers such that  $x + y + z \neq 0$ . Find the minimum value of

$$\frac{|x| + |x + 4y| + |y + 7z| + 2|z|}{|x + y + z|}.$$

5. Let  $ABCD$  be a convex quadrilateral with  $AB = 5$ ,  $AD = 17$ , and  $CD = 6$ . If the angle bisectors of  $\angle BAD$  and  $\angle ADC$  intersect at the midpoint of  $BC$ , find the area of  $ABCD$ .
6. (a) Find the smallest number of lines drawn on the plane so that they produce exactly 2022 points of intersection. (Note: For 1 point of intersection, the minimum is 2; for 2 points, minimum is 3; for 3 points, minimum is 3; for 4 points, minimum is 4; for 5 points, minimum is 4, etc.)  
(b) What happens if the lines produce exactly 2023 intersections?

### Test 2

1. Given a  $24 \times 24$  square grid, initially all its unit squares are coloured white. A *move* consists of choosing a row, or a column, and changing the colours of all its unit squares, from white to black, and from black to white. Is it possible that after finitely many moves, the square grid contains exactly 574 black unit squares?
2. Given  $\triangle ABC$ ,  $\angle CAB = 75^\circ$  and  $\angle ACB = 45^\circ$ .  $BC$  is extended to  $T$  so that  $BC = CT$ . Let  $M$  be the midpoint of the segment  $AT$ . Find  $\angle BMC$ .

3. Let  $n \geq 4$  be a positive integer. Consider any set  $A$  formed by  $n$  distinct real numbers such that the following condition holds: for every  $a \in A$ , there exist distinct elements  $x, y, z \in A$  such that  $|x - a|, |y - a|, |z - a| \geq 1$ . For each  $n$ , find the greatest real number  $M$  such that

$$\sum_{a \in A} |a| \geq M$$

always holds.

4. A two digit number  $s$  is *special* if  $s$  is the common two leading digits of the decimal expansion of  $4^n$  and  $5^n$ , where  $n$  is a certain positive integer. Given that there are two special numbers, find these two special numbers.

### Test 3

1. Let  $(a_n)_{n \geq 1}$  be a sequence of positive real numbers with the property that

$$(a_{n+1})^2 + a_n a_{n+2} \leq a_n + a_{n+2}$$

for all positive integers  $n$ . Show that  $a_{2022} \leq 1$ .

2. A number is called *Norwegian* if it has three distinct positive divisors whose sum is equal to 2022. Determine the smallest Norwegian number.

(Note: The total number of positive divisors of a Norwegian number is allowed to be larger than 3.)

3. Let  $ABCD$  be a cyclic quadrilateral. Assume that the points  $Q, A, B, P$  are collinear in this order, in such a way that the line  $AC$  is tangent to the circle  $ADQ$ , and the line  $BD$  is tangent to the circle  $BCP$ . Let  $M$  and  $N$  be the midpoints of  $BC$  and  $AD$ , respectively. Prove that the following three lines are concurrent: line  $CD$ , the tangent of circle  $ANQ$  at point  $A$ , and the tangent to circle  $BMP$  at point  $B$ .